



Modelling Financial Markets

Chapter 0: Financial Markets in 2026



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Acronyms used in this material (1/3)

- ▣ AI: Artificial Intelligence
- ▣ AMM: Automated Market Maker
- ▣ AP: Authorised Participant
- ▣ APT: Arbitrage Pricing Theory
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ ASE: Academia de Studii Economice din Bucureşti (the Bucharest University of Economic Studies)
- ▣ AUM: Assets Under Management
- ▣ BCBS: Basel Committee on Banking Supervision
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-FI: BET Financial Investment companies index
- ▣ BET-TR: BET Total Return (index)
- ▣ BIS: Bank for International Settlements
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori Bucureşti (the Bucharest Stock Exchange)
- ▣ CAGR: Compound Annual Growth Rate
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CBOE: Chicago Board Options Exchange
- ▣ CCP: Central CounterParty
- ▣ CDO: Collateralised Debt Obligation
- ▣ CFTC: Commodity Futures Trading Commission
- ▣ CIR: Cox–Ingersoll–Ross (model)
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DJIA: Dow Jones Industrial Average
- ▣ EBA: European Banking Authority
- ▣ EMH: Efficient Market Hypothesis

Acronyms used in this material (2/3)

- EOD: End of Day
- EODHD: EOD Historical Data (market-data provider)
- ES: Expected Shortfall
- ESMA: European Securities and Markets Authority
- ETF: Exchange-Traded Fund
- ETP: Exchange-Traded Product
- EU: European Union
- EVT: Extreme Value Theory
- EWMA: Exponentially Weighted Moving Average
- FRED: Federal Reserve Economic Data
- FRTB: Fundamental Review of the Trading Book
- FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- FX: Foreign eXchange
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GMM: Generalized Method of Moments
- GPT: Generative Pre-trained Transformer
- GPU: Graphics Processing Unit
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- HFT: High-Frequency Trading
- HJM: Heath–Jarrow–Morton (model)
- HML: High Minus Low (value factor)
- IBM: International Business Machines (Corporation)
- IPO: Initial Public Offering
- LLM: Large Language Model
- MC: Monte Carlo (simulation)
- MDD: Maximum DrawDown

Acronyms used in this material (3/3)

- ▣ MDE: Minimum Detectable Effect
- ▣ ML: Machine Learning
- ▣ MTF: Multilateral Trading Facility
- ▣ NAV: Net Asset Value
- ▣ NBER: National Bureau of Economic Research
- ▣ NISQ: Noisy Intermediate-Scale Quantum
- ▣ NYSE: New York Stock Exchange
- ▣ OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))
- ▣ OTC: Over-The-Counter
- ▣ QAE: Quantum Amplitude Estimation
- ▣ QR: Quick Response (code)
- ▣ ROL: Romanian old leu (ISO 4217 code, until 2005)
- ▣ RWA: Real-World Assets
- ▣ SD: Standard Deviation
- ▣ SE: Standard Error
- ▣ SEC: Securities and Exchange Commission
- ▣ SMB: Small Minus Big (size factor)
- ▣ SPDR: Standard & Poor's Depositary Receipts
- ▣ SR: Sharpe Ratio
- ▣ TARF: Target Accrual Redemption Forward
- ▣ UDR: Uzinele și Domeniile Reșița (the Reșița Works and Domains)
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VIX: CBOE Volatility Index
- ▣ VOC: Vereenigde Oostindische Compagnie (Dutch East India Company)

Chapter Route and Outcomes

Route of this chapter

- **Guiding question:** which financial comparisons can we trust, and how precisely do we know them?
- Part I: market structure, a short history of exchanges and financial models, the course data, measurement and its precision
- Part II: the 2000–2026 evidence, the Romanian market, digital assets, passive investing with a case study, one pass of the AI-assisted research loop
- Technical detail sits in the [Appendix](#) at the end

After this chapter you can (Seminar 0 exercise)

- compute simple and log returns, CAGR, annualised volatility and maximum drawdown of one asset on its own calendar (A1–A4, B1)
- audit a price series: state its price field and observation time, flag bad ticks and justify each cleaning decision (B6)
- join two assets' prices on common days, then estimate and interpret a change in their correlation between two periods (B3)
- report a mean return and a Sharpe ratio with a standard error and a 95% interval, and say what would make the interval narrower (A10, B1 Extended)
- spot a misleading empirical or AI-generated claim: missing uncertainty, an ignored rival explanation, a wrong formula or reference (C2)

Course Organisation

Programme and team

- ▣ Master's programme **Applied Statistics and Data Science**, Faculty of Cybernetics, Statistics and Economic Informatics, ASE
- ▣ Academic year 2026/2027
- ▣ Lecture: Prof. Daniel Traian Pele, danpele@ase.ro
- ▣ Seminar: Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

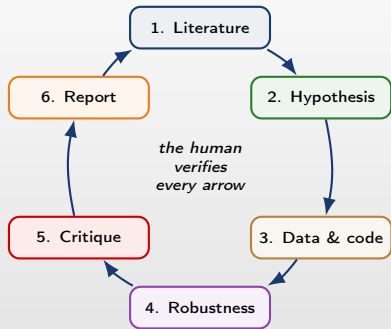
Grading

- ▣ Team project (GitHub, presentation, oral defence): **70%**
- ▣ Quizzes and seminar activity: **20%**; one quiz per chapter on the course website, 20 questions drawn from a bank, each answer explained
- ▣ Attendance: **10%**; at least **4 lectures** and **4 seminars**, recorded with the code of the day shown in the room (QR code or the Attendance form on the website)

The Team Project and Its Outputs

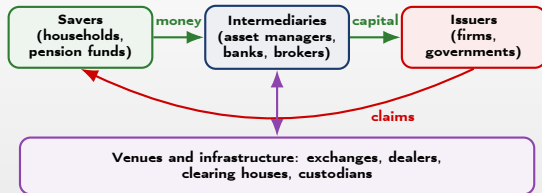
- ▣ Teams of 3–4 students, one research question on real market data; final presentation in [Chapter 19](#)
- ▣ One public GitHub repository per team
 - ▶ README (question, data, how to run), weekly commits that document the progress
 - ▶ every figure as a [Quantlet](#): one folder with code, `Metainfo.txt` (name, description, keywords, inputs, outputs) and the chart
- ▣ A result exists only if someone else can reproduce it from the repository
 - ▶ data loaded by code, every reported number produced by a script, random numbers with a fixed seed
- ▣ Tools: Python (pandas, numpy, scipy, statsmodels, scikit-learn, matplotlib), notebooks in Google Colab, the course repository github.com/danpele/MFM
- ▣ AI use is allowed and declared: every use in [AI_USE.md](#), every caught AI error in [AI_ERRORS.md](#)

The AI-Assisted Research Loop



- An AI assistant (an LLM such as Claude, ChatGPT or Gemini) can draft every step: literature map, competing hypotheses, code, robustness plans, a hostile referee report
- The human owns the question, the identifying assumption, the test and every reported number
- Verification checklist
 - ▶ every reference: the DOI (Digital Object Identifier) resolves and the title matches
 - ▶ every number: reproduced from the data by your own code
 - ▶ H_0 , test, sample and multiple-testing correction fixed before seeing results
- Typical LLM failures: invented references, plausible but wrong formulas, look-ahead in code, searching until something is “significant”

What Markets Do and Who Trades



- **This section answers:** who trades what, where, and how is the price we download formed?
- **Financial market:** a mechanism where buyers and sellers exchange claims on future cash flows (shares, bonds, currencies, contracts, tokens) at prices set by supply and demand
- **Functions:** **capital allocation**, **risk sharing**, **liquidity** (converting assets to cash quickly), **price discovery** (prices aggregate dispersed information), **payments and settlement**
- **Buy side:** retail and institutional investors, asset managers, **passive** (replicate an index) or **active** (try to beat it)
- **Sell side:** brokers execute orders, dealers trade from inventory, **market makers** quote a buy and a sell price

Asset Classes

Class	Examples	Cash flow	Main risks
Equities	shares, equity ETFs	dividends, residual claim	market, firm-specific
Fixed income	government and corporate bonds	coupons, principal	interest rate, credit
Money market	T-bills (< 1 year), repo (secured loan)	short-term interest	liquidity
FX	EUR/USD, EUR/RON	interest differential	currency, policy
Commodities	gold, oil futures	none (roll yield)	supply, storage
Derivatives	futures, options, swaps	contingent payoffs	leverage, counterparty
Digital assets	Bitcoin, Ether, stablecoins	none / protocol-specific	volatility, regulation

- **ETF** (exchange-traded fund): a fund whose shares trade like a stock; **FX**: the currency market, EUR/RON = lei per euro; **roll yield**: gain or loss from replacing an expiring futures contract
- One statistical toolbox for all of them: returns, volatility, dependence and tails (full list of terms in the [Appendix](#))

The Limit Order Book and a Market Order



- **Limit order book:** all waiting buy (**bid**) and sell (**ask**) orders, sorted by price; best bid 99.99, best ask 100.01
- **Bid-ask spread:** best ask minus best bid (0.02), the price of trading at once; **depth:** quantity near the best prices
- Order types: a resting **limit order** provides liquidity; a **market order** consumes it; a **stop order** becomes a market order at a trigger price, and clustered stops can accelerate a fall
- **Question for the room:** a market buy of 5 units arrives; at what average price is it executed, and what is the spread afterwards?

The Limit Order Book and a Market Order

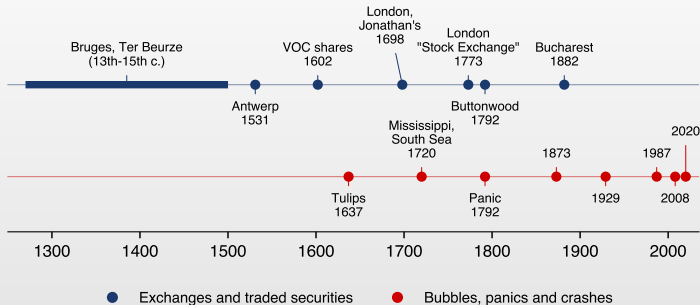


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- **Answer:** it consumes the book (orange)
 - ▶ 2 units at 100.01 and 3 at 100.02: average $(2 \times 100.01 + 3 \times 100.02)/5 = 100.016$
 - ▶ the best ask moves to 100.02 and the spread widens to 0.03: **market impact** (Chapter 10)

Observation Times and Price Noise

- A “daily price” is a price observed at a specific time
 - ▶ exchange-traded assets: the closing auction (S&P 500: 20:00 or 21:00 UTC); in a **call auction**, orders are collected and matched at one price
 - ▶ auctions versus continuous trading, and why speed is rewarded: ▶ [Appendix](#)
 - ▶ Bitcoin: no close; the price of day t is taken at 24:00 UTC, the end of day t
 - ▶ EUR/RON: the BNR reference rate, fixed around 13:00 Bucharest time
- The venue decides what is recorded: exchanges, OTC (over-the-counter) dealers, dark pools and 24/7 crypto platforms show different prices and trades (venue table in the ▶ [Appendix](#))
- Different observation times split the same news across two days: correlations of assets with different closes are attenuated (▶ [Appendix](#))
- **Bid–ask bounce**: trades alternate between bid and ask, so observed price changes are negatively autocorrelated even when the efficient price is a random walk
 - ▶ the Roll model gives $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -s^2/4$ for a spread s ([Roll \(1984\)](#); derivation in the ▶ [Appendix](#))
- Spreads, impact and fragmentation add noise to observed prices, most visible at high frequency ([Chapters 9 and 10](#))

Five Centuries of Exchanges and Crashes



- **This section answers:** how did exchanges, crashes and models grow together, and what did each wave of models add?
- Exchanges grew where merchants met regularly; shares traded continuously from the 1600s, and bubbles and crashes followed almost at once
- The Bucharest Stock Exchange (1882) has its own frames in the [Appendix](#)

The First Exchanges: from Bruges to New York



The Amsterdam exchange of Hendrick de Keyser, 1612

Engraving: Claes Jansz. Visscher (1612); public domain;

Stadsarchief Amsterdam, Wikimedia Commons

- **Bruges**, 13th–15th centuries: merchants met on the square before the inn of the Van der Beurse family; the name became *bourse*, *beurs*, *bursă* (Gelderblom (2013))
- **Antwerp**, 1531: a building purpose-built for daily trading opens (Gelderblom (2013))
- **Amsterdam**, 1602: the Dutch East India Company, VOC (Vereenigde Oostindische Compagnie), issues transferable shares
 - ▶ continuous trading, forward contracts, options and short selling within a decade (Gelderblom & Jonker (2004); Petram (2014))
- **London**: from 1698 John Castaing publishes price lists at Jonathan's Coffee House; in 1773 the brokers call their new building the Stock Exchange (Neal (1990))
- **New York**, 17 May 1792: 24 brokers sign the Buttonwood Agreement, the origin of the NYSE (New York Stock Exchange) (Sylla, Wright & Cowen (2009))

The First Bubbles, Panics and Crashes

- **Tulip mania**, Holland 1636–37: prices of bulb contracts rose many-fold, then collapsed in February 1637
 - ▶ later research shows a smaller episode than the legend: rare bulbs lost value like any new flower variety ([Garber \(1989\)](#)); few bankruptcies, little damage ([Goldgar \(2007\)](#)); more in the [Appendix](#)
- **1720**: the Mississippi Company in Paris (John Law) and the South Sea Company in London rise and collapse within months
 - ▶ prices rose most for firms tied to the new Atlantic trade and to insurance: a bubble around a real innovation ([Frehen, Goetzmann & Rouwenhorst \(2013\)](#))
- **Panic of 1792**, New York: the Treasury under Alexander Hamilton acts as a lender of last resort ([Sylla, Wright & Cowen \(2009\)](#))
- **1873**: crashes in Vienna (May) and New York (September); the NYSE closes for ten days and the US contraction dated by the NBER (National Bureau of Economic Research) lasts until 1879 ([NBER business-cycle dates](#))
- **1929**: Black Tuesday, in the [Appendix](#)
- **19 October 1987**: the DJIA (Dow Jones Industrial Average) falls 22.6% in one day, with large falls in every major market ([Roll \(1988\)](#))
- Recurring ingredients: a new asset or technology, credit and leverage, then forced selling ([Chapter 17](#))

Financial Models (1/4): Are Prices Predictable, and Which Risk Is Paid?



Louis Bachelier; William F. Sharpe,
2007

Bachelier: unknown author; public
domain; Sharpe: Larry D. Moore
(2007); CC BY 4.0; Wikimedia
Commons

- **Problem:** can past prices predict future prices, and which risks should earn a reward?
- **Random walk and efficiency**
 - ▶ Bachelier (1900): bond prices follow a Brownian motion; Mandelbrot (1963): price changes have far heavier tails than the Normal distribution allows
 - ▶ EMH (efficient market hypothesis, Fama (1970)): prices fully reflect available information
- **Portfolios and equilibrium**
 - ▶ Markowitz (1952): choose weights by mean and variance; diversification depends on covariances
 - ▶ CAPM (capital asset pricing model, Sharpe (1964)): only market risk is priced
- **Made possible by:** probability theory, long price series, the first stock-return databases; in this course: Chapters 1–4
- Details with formulas: random walks and efficient markets (▶ Appendix), portfolios, CAPM and factor models (▶ Appendix) and all model families on one timeline (▶ Appendix)

Financial Models (2/4): How to Price Options and Measure Risk?



Fischer Black, Myron Scholes, Robert C.
Merton

Black: Dalmatine; public domain; Scholes:

Nobellaureatesphotographer (2008); CC

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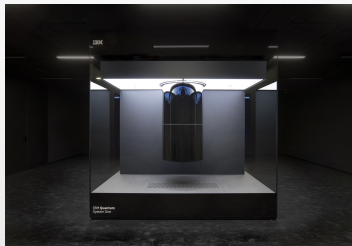
domain; Wikimedia Commons

- **Problem:** what is an option worth, and how much capital should a bank hold against market risk?
- **Derivatives:** a continuously rebalanced hedge replicates an option, so its price does not depend on the stock's expected return (Black & Scholes (1973); Merton (1973))
- **Volatility:** ARCH and GARCH make today's variance depend on past shocks, so volatility clusters and can be forecast (Engle (1982); Bollerslev (1986))
- **Risk measurement:** banks' capital from VaR 1% (the loss exceeded with probability 1%), later ES 2.5% (the mean loss in the worst 2.5% of cases) (BCBS (1996); BCBS (2019))
- **Made possible by:** stochastic calculus, listed options (1973), daily and intraday data, bank regulation; in this course: Chapters 5–12
- Lesson of 2008: one Gaussian-copula parameter priced every tranche of a CDO (collateralised debt obligation, paid from a pool of loans), a model risk (▶ Appendix)
- Details: interest-rate and stochastic-volatility models (▶ Appendix), volatility models (▶ Appendix) and coherent risk measures (▶ Appendix)

Financial Models (3/4): What Can Be Learned from Big Data and Text?

- ▣ **Problem:** hundreds of firm signals and millions of documents do not fit a small parametric model
- ▣ **Machine learning (ML):** flexible functions fitted to data and selected by out-of-sample performance rather than by theory
 - ▶ [Gu, Kelly & Xiu \(2020\)](#): trees and neural networks beat linear models on about 30,000 US stocks and 94 firm characteristics; the monthly out-of-sample R^2 stays below 1%, small but economically valuable
- ▣ **Generative AI:** the Transformer ([Vaswani et al. \(2017\)](#)) underlies today's LLMs (large language models)
 - ▶ [Lopez-Lira & Tang \(2026\)](#): ChatGPT scores of news headlines predict next-day stock returns out of sample
 - ▶ foundation models for time series forecast new series without retraining ([Ansari et al. \(2024\)](#))
- ▣ **Made possible by:** large panels, cheap computing, GPU (graphics processing unit) hardware, text at scale; in this course: [Chapters 13–15](#)
- ▣ Main risks: overfitting, look-ahead (the model may have read texts written after the forecast date), many trials on the same data
- ▣ Details: neural pricing, deep hedging and the out-of-sample R^2 (▶ [Appendix](#))
- ▣ Details: BloombergGPT, Chronos and open problems (▶ [Appendix](#))

Financial Models (4/4): Quantum Computing, in Simple Words



IBM Quantum System One, Ehningen
(Germany), 2021

Photo: IBM Research (2021); CC BY 2.0;
Wikimedia Commons

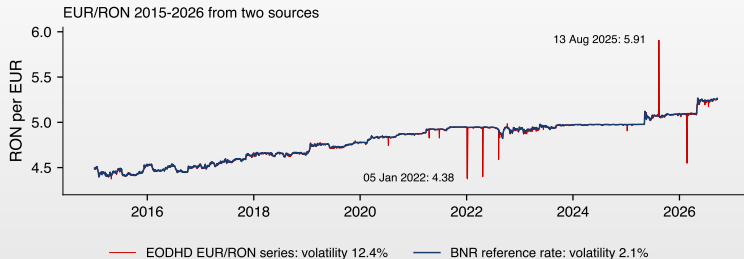
- **Problem:** Monte Carlo pricing and risk need millions of simulated scenarios
- **Idea:** a quantum computer works with qubits, which can hold a combination of 0 and 1 at once; interference makes the right answer more likely (Orús, Muga & Lizaso (2019); Herman et al. (2023))
- **Proven gain:** for an error ε a classical simulation needs about $1/\varepsilon^2$ samples, a quantum algorithm about $1/\varepsilon$ steps: 10^8 against 10^4 for $\varepsilon = 10^{-4}$ (Montanaro (2015)); a quadratic, not an exponential, gain
- **Not yet practical:** today's qubits are noisy; a useful derivative-pricing run needs about 8 000 error-corrected qubits (Chakrabarti et al. (2021)), far beyond current machines
- No wave replaced the earlier ones: GARCH still feeds VaR, factor models still benchmark ML (summary table in the [Appendix](#))
- Details: the promise ([Appendix](#)), the hardware limits ([Appendix](#)) and the calculation step by step ([Appendix](#))

Data Used in This Course

Data	Content	Format
Daily market data from EODHD (EOD Historical Data), saved in the course repository	indices, stocks, ETFs, FX, crypto, yields	CSV
FRED (Federal Reserve Economic Data)	US rates, exchange rates, macro series	public CSV
BNR	official RON reference exchange rates	public XML
DefiLlama	stablecoin supply, DeFi (decentralised finance)	public JSON

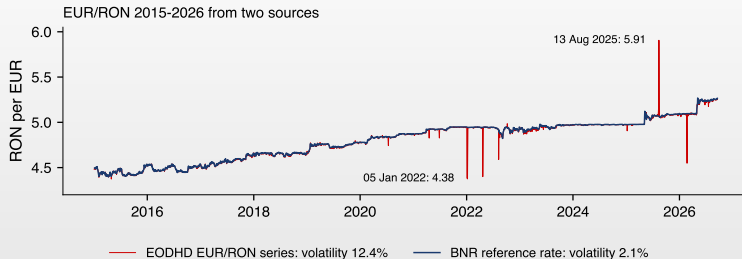
- ▣ **This section answers:** which data feed every chapter, and which price field and calendar should you use?
- ▣ Stocks and ETFs: adjusted close (dividends, splits); indices, FX and crypto: close; bond yields: the closing quoted yield, not a bond price
- ▣ One series: returns on its own calendar; joint analysis: join the *prices* on common days, then take returns (Friday → Monday for both)
- ▣ Annualise with the observed frequency: 365 for crypto, about 260 for spot gold on weekdays, 252 for equities
- ▣ **Survivorship:** samples rebuilt from today's survivors, without delisted firms or dead coins, make past returns look too good ([Brown, Goetzmann, Ibbotson & Ross \(1992\)](#))

Data Pitfalls: a Suspicious EUR/RON Move



- **Question for the room:** EUR/RON from EODHD (aggregated interbank quotes, not an official rate) jumps by 15% on 13 Aug 2025 and falls back the next day; market event or data error, and how would you decide?

Data Pitfalls: a Suspicious EUR/RON Move



- ❑ **Question for the room:** EUR/RON from EODHD (aggregated interbank quotes, not an official rate) jumps by 15% on 13 Aug 2025 and falls back the next day; market event or data error, and how would you decide?
- ❑ **Answer:** a data error, a **bad tick**: the jump reverses at once and the BNR reference rate, a second source, does not move (also $\pm 12\%$ in early Jan 2022)
 - ▶ annualised volatility 2015–2026: 12.4% with these ticks, 2.1% with the BNR rate; EUR/RON always from BNR
 - ▶ Romania 10-year yield: a 0.00% print on 24 Dec 2013 and stale values on Sundays in late 2022

Returns, Growth and Annualisation

- ▣ **This section answers:** how do we measure growth and risk, and how precisely do we know the answer?
- ▣ **Price index** (S&P 500, BET) ignores dividends; **total-return index** (BET-TR) and **adjusted close** (stocks, ETFs) reinvest them
- ▣ Simple return $R_t = P_t/P_{t-1} - 1$; log return $r_t = \ln(P_t/P_{t-1}) = \ln(1 + R_t)$, additive over time
- ▣ **CAGR** (compound annual growth rate) $= (P_{\text{end}}/P_{\text{start}})^{1/Y} - 1$, with $Y = \text{calendar days}/365.25$
- ▣ **Annualised volatility** $\hat{\sigma}_d\sqrt{q}$: $\hat{\sigma}_d$ = standard deviation of daily log returns, q = observed return intervals per year (about 252 for equities, 365 for crypto)
 - ▶ exact only for uncorrelated returns with constant variance; the variance ratio and when $\sqrt{h}$ scaling fails: [▶ Appendix](#)
- ▣ Worked example: S&P 500, 2 Jan 2015 – 18 Sep 2026 (4 277 days, $Y = 11.71$)
 - ▶ multiple $\times 3.72$: $\text{CAGR} = 3.72^{1/11.71} - 1 = e^{\ln 3.72/11.71} - 1 = 11.9\%$
 - ▶ with dividends reinvested (SPY, the SPDR S&P 500 ETF): 13.7% a year; the 1.8 percentage point (pp) a year difference comes mainly from reinvested dividends, net of the ETF's fees and tracking differences
- ▣ Notation for drawdowns, correlations, yields and spreads, used in the next sections: [▶ Appendix](#)

How Precisely Do We Know an Expected Return?

- Daily simple returns R_t independent and identically distributed (iid), with mean μ_d and standard deviation σ_d ; $T = qY$ observations over Y years
- Annualised mean $\hat{\mu} = q\bar{R}$ has standard error

$$\text{SE}(\hat{\mu}) = \frac{q\sigma_d}{\sqrt{T}} = \frac{\sigma_a}{\sqrt{Y}}, \quad \sigma_a = \sigma_d\sqrt{q}$$

- S&P 500 with dividends (SPY), 2015–2026: $\hat{\mu} = 14.4\%$, $\sigma_a = 17.5\%$ (from simple returns; 17.6% from log returns), $Y = 11.71$
 - ▶ $\text{SE} = 17.5\%/\sqrt{11.71} = 5.1 \text{ pp}$; 95% confidence interval [4.3%, 24.4%]
- **Question for the room:** to halve this SE, would you rather have four times as many observations (hourly instead of daily data) or four times as many years?

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- **Answer:** four times as many years (47): the SE depends only on the **span** Y , not on the sampling frequency (Merton (1980))

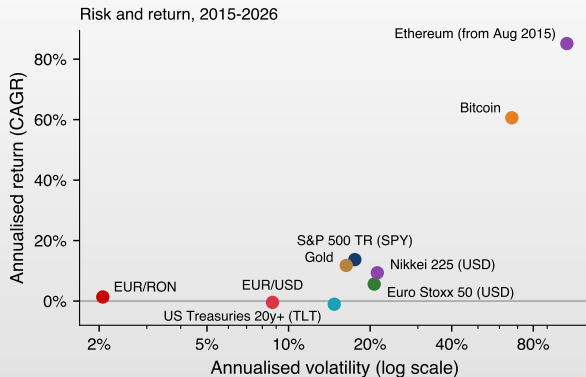
The Sharpe Ratio Is an Estimate

- ▣ **Sharpe ratio (SR)**, risk-free rate set to 0: $SR_d = \mu_d / \sigma_d$ per day (mean and SD of daily simple returns), annualised $SR_a = \sqrt{q} SR_d$
- ▣ For iid returns following the Normal distribution (Lo (2002)):

$$SE(\widehat{SR}_a) = \sqrt{\frac{1 + SR_a^2 / (2q)}{Y}} \approx \frac{1}{\sqrt{Y}}$$

- ▣ SPY, 2015–2026: $\widehat{SR}_a = 0.82$, $SE = 0.29$, 95% confidence interval [0.25, 1.39]
- ▣ Like the mean, the SR is known only to about $\pm 2 / \sqrt{Y}$: 11.7 years cannot separate 0.8 from 0.5
- ▣ The delta-method derivation, and how fat tails, dependence and picking the best of many strategies change the SE: [► Appendix](#)

Risk and Return, 2015–2026



- **This section answers:** what did markets deliver since 2000, and which differences can we trust?
- Vertical axis: CAGR, the **realised compound growth**; horizontal: annualised volatility of daily log returns, with the observed q ; all in USD, except EUR/RON (RON per EUR)
- S&P 500 with dividends: 13.7% a year at 17.6% volatility; gold 11.8% at 16.3% (weekday quotes); Bitcoin 60.6% at 66.7%
- The next slide tests two other targets: the **arithmetic mean return** (an expected return) and the **Sharpe ratio** (risk-adjusted); its intervals do not cover the plotted CAGR
- In the [Appendix](#): the growth of 1 USD for each asset since 2015

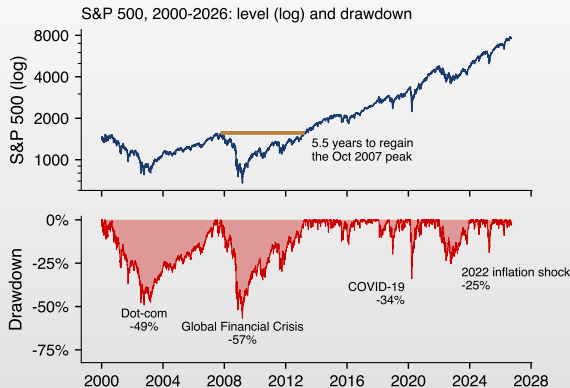
Is Any Ranking in the Risk–Return Chart Significant?

- Standard error of an annualised mean return, $\sigma_a/\sqrt{Y}$ with $Y = 11.7$
 - ▶ S&P 500 with dividends 5.1 pp, gold 4.8 pp, Euro Stoxx 50 6.0 pp, Nikkei 225 6.2 pp, Bitcoin 19.4 pp
 - ▶ S&P 500 minus gold on common days: +1.9 pp a year, SE 6.8 pp, $t = 0.28$
- Sharpe ratios: Bitcoin 1.05, S&P 500 0.82, gold 0.77, Nikkei 225 0.53, Euro Stoxx 50 0.37, long Treasuries (TLT, iShares 20+ Year Treasury Bond ETF) 0.00, each with SE ≈ 0.29
- Overlapping intervals do not test a difference: bootstrap the paired difference on common days (Seminar 0, B1 Extended)
 - ▶ S&P 500 minus gold: +0.06, 95% interval $[-0.65, +0.81]$
 - ▶ S&P 500 minus Euro Stoxx 50: +0.46, $[0.09, 0.85]$, significant alone; none of the 10 pairs of non-crypto assets survives a Bonferroni correction (each test at 0.5%)
- **Question for the room:** the S&P 500 minus gold interval contains zero; does this show that the two Sharpe ratios are equal?

Is Any Ranking in the Risk–Return Chart Significant?

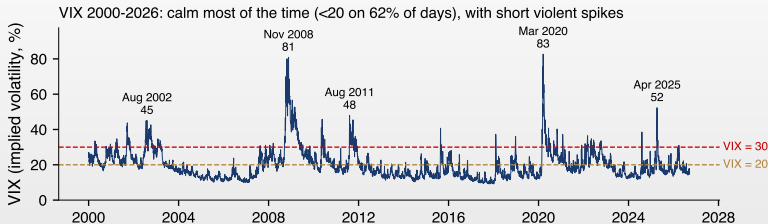
- Standard error of an annualised mean return, $\sigma_a/\sqrt{Y}$ with $Y = 11.7$
 - ▶ S&P 500 with dividends 5.1 pp, gold 4.8 pp, Euro Stoxx 50 6.0 pp, Nikkei 225 6.2 pp, Bitcoin 19.4 pp
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- **Question for the room:** the S&P 500 minus gold interval contains zero; does this show that the two Sharpe ratios are equal?
- **Answer:** no; the data are compatible with gold ahead by 0.65 and with the S&P 500 ahead by 0.81: the comparison is inconclusive, not evidence of equality
- The chart describes the sample; it does not rank expected returns

Crises and Drawdowns



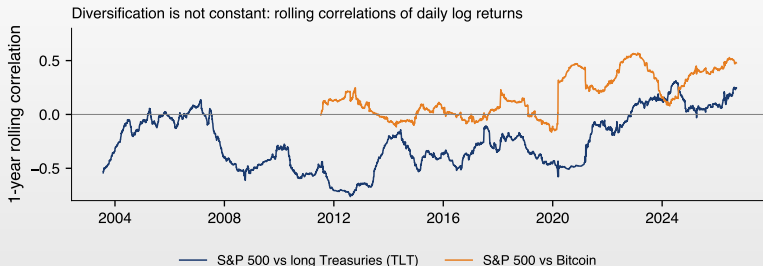
- Lower panel: $DD_t = P_t / \max_{s \leq t} P_s - 1$ of the S&P 500 (price index)
- Four troughs since 2000: dot-com -49% (Oct 2002), global financial crisis -57% (Mar 2009), COVID-19 -34% (Mar 2020, the fastest), inflation shock -25% (Oct 2022)
- 1 565.15 (9 Oct 2007) to 676.53 (9 Mar 2009); the peak was regained only on 28 Mar 2013
- The MDD deepens with the sample length: compare MDDs only over the same horizon ([Appendix](#))

Volatility Regimes: the VIX



- **VIX** (CBOE Volatility Index): the S&P 500 volatility over the next 30 days **implied** by option prices, in % per year
- Since 2000: below 20 on 62% of days, above 30 on 9%; peaks 82.7 (16 Mar 2020), 80.9 (20 Nov 2008), 52.3 (8 Apr 2025)
- Long calm periods, short intense stress: volatility comes in regimes ([Chapters 5 and 9](#))
- The VIX exceeded the volatility realised afterwards on 83% of days: the variance risk premium, in the [Appendix](#)

Diversification Benefits Change over Time



- 252-day rolling correlations of daily log returns, prices joined on common days first
- Stocks vs long Treasuries: mean rolling -0.43 (2010–2020), $+0.08$ (2022–2026); whole-period correlations -0.47 and $+0.11$, a different estimator
- Stocks vs Bitcoin: mean rolling 0.01 (2016–2019), 0.38 (2022–2026); fixed-correlation portfolio models miss such shifts (Chapters 4 and 6)
- Two traps for crypto–equity correlations, asynchronous closes and the volatility bias: [Appendix](#)

Has the Stock–Bond Correlation Really Changed?

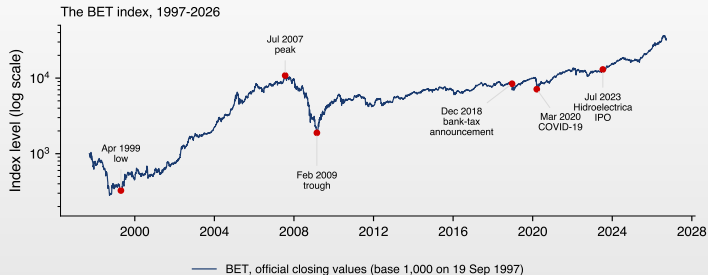
- ▣ Whole-period correlation S&P 500 – TLT: -0.47 in 2010–2020 ($n = 2\,769$), $+0.11$ in 2022–2026 ($n = 1\,182$); estimated change $+0.58$
- ▣ Fisher z test, iid pairs from a bivariate Normal distribution: $\operatorname{atanh} \hat{\rho}$ has variance about $1/(n - 3)$
 - ▶ $z = 17.9$, $p < 10^{-10}$; iid 95% interval for the change $[0.52, 0.65]$
- ▣ Volatility clustering makes pairs dependent and fat tails break normality: stationary bootstrap, mean block 20 days
 - ▶ 95% interval $[0.49, 0.68]$, 1.54 times wider; the change survives (Seminar 0, B3 Extended)
- ▣ The test establishes a change in association between two periods, not its cause
 - ▶ one candidate cause, the 2022 inflation and rate shock, needs its own test; rates and the yield curve:

▶ Appendix

The Bucharest Stock Exchange Today

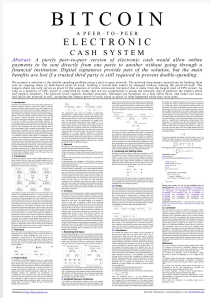
- ▣ **This section answers:** what do the same measurements show on the market closest to us?
- ▣ **BVB** (Bursa de Valori București): opened in 1882, closed in 1948, re-established in 1995 (history in the [Appendix](#))
- ▣ **BET** (Bucharest Exchange Trading index): the most traded Romanian companies, launched on 19 Sep 1997 at 1 000 points [BVB \(indices\)](#); **BET-TR**: its total-return version, with reinvested dividends
- ▣ Largest companies: banks (Banca Transilvania, BRD), energy (OMV Petrom, Romgaz, Hidroelectrica), utilities (Transgaz)
- ▣ Classified as **Secondary Emerging** by [FTSE Russell \(2026\)](#) since September 2020
- ▣ Milestones: Hidroelectrica, the largest IPO (initial public offering) in BVB history, EUR 1.9 bn, July 2023 [BVB \(history\)](#); **TVBETETF**, an ETF on BVB that replicates BET

The BET Index, 1997–2026: a Long Recovery



- Official closing values, log scale; from the peak of 24 Jul 2007 to the trough of 25 Feb 2009: -82.5% , regained only on 16 Mar 2021, after **13.6 years**
- 19 Dec 2018: -11.2% in one day, after the bank-tax announcement; 18 Sep 2026: $\times 32.8$ since launch, CAGR 12.8% over 29 years
- Since 2015: dividend-adjusted equities ([▶ Appendix](#)), the leu ([▶ Appendix](#)) and Romanian versus German government yields ([▶ Appendix](#))

Bitcoin, Ethereum and Tokens



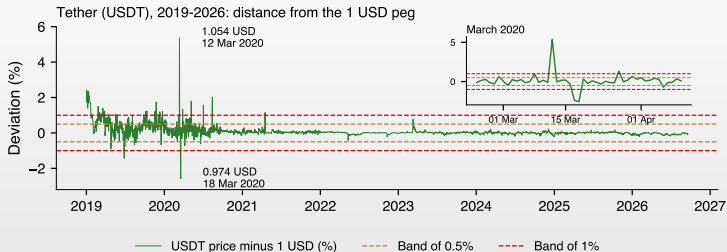
Bitcoin white paper (2008)

Text: [Satoshi Nakamoto \(2008\)](#); **poster:**

[DailyCoinPost \(2018\)](#); [CC0](#); [Wikimedia Commons](#)

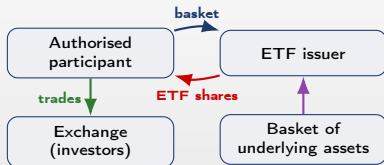
- **This section answers:** what are crypto assets and stablecoins, and what do their prices measure?
- 31 Oct 2008: *Bitcoin: A Peer-to-Peer Electronic Cash System*, by the pseudonymous Satoshi Nakamoto [Nakamoto \(2008\)](#)
- **Blockchain:** a shared ledger of transactions, copied on many computers and extended in blocks, without a trusted intermediary
- **Bitcoin (2009):** fixed supply rule (at most 21 million coins); **proof of work:** adding a block costs computing power, which makes rewriting history expensive
- **Ethereum:** a blockchain that runs programs (**smart contracts**); its coin is Ether; a **token** is a unit on a blockchain representing a coin, a claim or a right
- [Härdle, Harvey & Reule \(2020\)](#): statistical properties, pricing and regulation; [Pele et al. \(2023\)](#): tail, memory and moment factors set cryptos apart from other asset classes
- Bitcoin is still about 3.2 times as volatile as the S&P 500 ([Appendix](#))
- Since January 2024 spot Bitcoin ETFs trade in the US ([Appendix](#)); how ETFs work: next section

Stablecoins and the Peg



- ▣ **Stablecoin**: a token designed to keep a fixed value (**peg**), usually 1 USD; **reserve-backed** (cash, T-bills, redeemable at 1 USD) or **algorithmic** (a second token and trading rules); **depeg**: the price leaves the peg
- ▣ Tether (USDT), 2019–2026: within $\pm 0.5\%$ of 1 USD on 93% of days, within $\pm 1\%$ on 98%; the algorithmic TerraUSD lost its peg and collapsed in May 2022
- ▣ Supply is not the peg: the circulating value can fall while the price stays at 1 USD (supply chart in the [Appendix](#))
- ▣ Tokenised real-world assets and the EU and US rules for crypto assets: [Appendix](#)

ETFs: Creation and Redemption



- **This section answers:** how does an ETF keep its price near its assets, and what does cap-weighting imply for investors?
- Most ETFs are **passive**: they replicate an index at low cost
- **Net asset value (NAV)**: the fund's assets minus its liabilities, per share, computed once a day
- **Authorised participant (AP)**: a large dealer allowed to deal directly with the issuer
- **Creation**: the AP delivers the basket and receives new ETF shares; **redemption**: the reverse
- This two-way channel keeps the market price close to the NAV; ETFs now hold equities, bonds, commodities and, in the US since January 2024, spot Bitcoin

Worked Example: Arbitrage Keeps the ETF Price Near NAV

- ▣ Illustrative numbers: $NAV = 100$, the ETF trades at 100.50; **premium** $(P - NAV)/NAV = 0.5\%$
- ▣ **Question for the room:** which trade does an authorised participant make, and what happens to the premium?

Worked Example: Arbitrage Keeps the ETF Price Near NAV

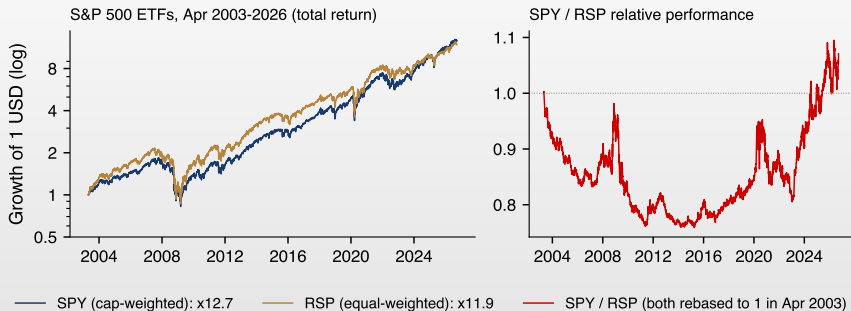
- ▣ Illustrative numbers: $NAV = 100$, the ETF trades at 100.50; **premium** $(P - NAV)/NAV = 0.5\%$
- ▣ **Question for the room:** which trade does an authorised participant make, and what happens to the premium?
- ▣ **Answer:** the AP buys the basket for 100, delivers it to the issuer, receives one new ETF share and sells it for 100.50
 - ▶ profit 0.50 per share before costs; the extra ETF shares push the price down until the premium disappears
 - ▶ at a **discount** ($P < NAV$) the AP buys ETF shares and redeems them for the basket
- ▣ Premiums persist where the NAV is stale or the basket is illiquid, e.g. foreign stocks or bonds (Petajisto (2017))
- ▣ The speed at which premiums decay can be measured with an AR(1) half-life (▶ [Appendix](#))

Cap-Weighted versus Equal-Weighted Indices

Definition 1 (Index weights and effective diversification)

- Capitalisation weights $w_i = MC_i / \sum_{j=1}^n MC_j$, market capitalisation $MC_i = \text{price} \times \text{number of shares}$; equal weights $w_i = 1/n$, rebalanced periodically
- Effective number of stocks $N_{\text{eff}} = 1 / \sum_i w_i^2$ (inverse Herfindahl): n for equal weights, far fewer when a few firms dominate
- SPY**: the S&P 500 with capitalisation weights; **RSP** (Invesco S&P 500 Equal Weight ETF): the same stocks, each $1/500$
- A cap-weighted index is less diversified than its number of stocks suggests; its risk depends on a few firms and themes ([Chapters 3, 4 and 17](#))
- Passive cap-weighted funds hold each firm in proportion to its size, so their investors carry the concentration of the largest firms

Cap-Weighted versus Equal-Weighted Performance: SPY and RSP



- Total return (adjusted close) since RSP's launch in April 2003: SPY $\times 12.7$, RSP $\times 11.9$; the ratio ends at 1.07
- SPY/RSP rose by 31% since January 2023: a handful of very large companies drove the index
- The ratio measures relative performance; it is not a direct measure of concentration or of passive ownership

Case Study: What Happens When Passive Investing Grows?

- **This section answers:** what does a landmark paper find about the effect of passive investing on prices?
- Haddad, Huebner & Loualiche (2025), *American Economic Review* 115(3), 975–1018
- **Question:** when some investors turn passive, do the others trade more aggressively and undo the effect on prices?
- **Demand elasticity** $\mathcal{E}_i$: the percentage fall in the shares investor i demands after a 1% price rise; a passive investor has $\mathcal{E}_i = 0$
- **Aggregate elasticity** $\mathcal{E}_{agg}$: the holdings-weighted average of the $\mathcal{E}_i$ (eq. 2); a demand shift $\Delta \bar{d}$ moves the log price by $\Delta p = \mathcal{E}_{agg}^{-1} \Delta \bar{d}$ (eq. 3)
 - ▶ lower $\mathcal{E}_{agg}$: prices more volatile and less informative (Section II.A)
- Link: stocks with higher ETF ownership are more volatile (Ben-David, Franzoni & Moussawi (2018))

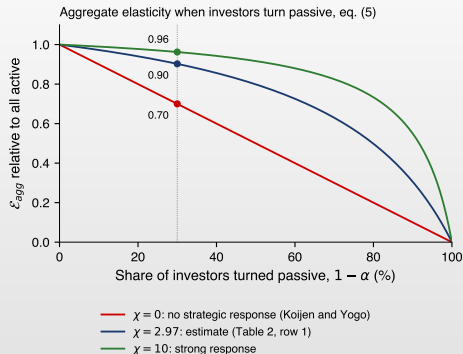
Case Study: an Illustrative Scenario, 30% Turn Passive

- **Strategic response** χ : $\mathcal{E}_i = \bar{\mathcal{E}}_i - \chi \mathcal{E}_{agg}$ (eq. 4), with $\bar{\mathcal{E}}_i$ the investor's own component; $\chi = 0$: nobody reacts; $\chi \rightarrow \infty$: the others fully offset any change
- Eq. (5): $\mathcal{E}_{agg}$ after a share $1 - \alpha$ of identical investors turns passive, relative to all active:

$$\alpha + (1 - \alpha) \frac{\chi^\alpha}{1 + \chi^\alpha}$$

- **Question for the room**: 30% turn passive ($\alpha = 0.7$) and nobody reacts ($\chi = 0$); what share of the aggregate elasticity is left?

Case Study: an Illustrative Scenario, 30% Turn Passive



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$$\alpha + (1 - \alpha) \frac{\chi \alpha}{1 + \chi \alpha}$$

- **Question for the room:** 30% turn passive ($\alpha = 0.7$) and nobody reacts ($\chi = 0$); what share of the aggregate elasticity is left?
- **Answer:** 0.70; with the paper's $\chi = 2.97$, 0.90: two-thirds of the direct fall is offset; $\chi = 10$ gives 0.96
- At 100% passive all curves reach 0: nobody is left to absorb demand shocks

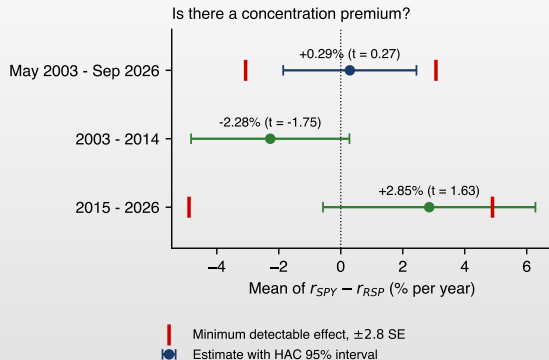
Case Study: Estimate, Identification and One Limitation

- **Data:** quarterly holdings of US institutional managers (SEC Form 13F), 2001–2020, about 80% of US market capitalisation
- **Estimate:** strategic response $\chi = 2.97$ (SE 0.47; Table 2, row 1), so a 95% interval of about $[2.0, 3.9]$: investors react, far from both 0 and ∞
- **Identifying assumption:** the same investor meets a different mix of other investors in each stock, which separates its own elasticity from χ (Theorem 1)
 - ▶ instruments, GMM (generalised method of moments) and the full demand system: ▶ Appendix
- **Implication:** the median active share fell from 81.4% (2001) to 59.5% (2020), $\ln(81/59) = 32\%$
 - ▶ pass-through to $\mathcal{E}_{agg}$ (eq. 28) $1/(1 + 3 \times 0.68) = 33\%$: stocks are about 11% less elastic; inelastic stocks are more volatile and less liquid (Table 5)
 - ▶ robustness and reduced-form checks: ▶ Appendix
- **One limitation:** one χ for all investors, with no fund flows (Section I.B); all stated limits and what the course data can check: ▶ Appendix
- **Open question:** does passive investing change price formation? [Sammon \(2025\)](#): higher passive ownership makes prices anticipate earnings announcements less; passive ownership is endogenous

Hypothesis: Is There a Concentration Premium?

- **This section answers:** how does one pass of the AI-assisted research loop look on a new question?
- A separate descriptive exercise: comparing SPY with RSP does not test, and cannot identify, the case study's passive-ownership mechanism
- A typical assistant proposal: "passive flows favour the largest firms, so cap weights beat equal weights"
- Discovery loop in one pass
 - ▶ **literature:** ask for peer-reviewed papers with DOIs, then resolve each DOI yourself
 - ▶ **hypothesis:** three competing explanations (passive flows, sector composition, earnings growth of the largest firms), each with one implication the others do not share
 - ▶ **data and code:** SPY and RSP adjusted (total-return) prices, joined on common days; first replicate $\text{SPY} \times 12.7$ and $\text{RSP} \times 11.9$
- **Test:** $d_t = r_{\text{SPY},t} - r_{\text{RSP},t}$ (daily log returns); $H_0 : \mathbb{E}[d_t] = 0$ against $H_1 : \mathbb{E}[d_t] \neq 0$
- Checks fixed in advance: the price type, the calendar join, the windows (full sample and two halves) and the HAC (heteroskedasticity and autocorrelation consistent) Newey–West t -statistic

Mini-Case: Estimate, Interval and MDE



- May 2003 – September 2026 (5 884 days): mean spread +0.29% a year, $t = 0.27$; halves fixed in advance: -2.28% ($t = -1.75$), then +2.85% ($t = 1.63$)
- Dots: annualised mean of d_t ; bars: HAC 95% intervals; red ticks: $\pm$ MDE
- **Minimum detectable effect (MDE)**: the smallest true effect detected with 80% power by a two-sided 5% test, about $2.8 \times \text{SE}$: 3.1% a year for the full sample
- Every interval contains 0 and the sign flips between the halves: the “premium” is a feature of the chosen window
- “Not significant” is not “no effect”; report every window tried
- Formula, assumptions and the causal caveat: [Appendix](#)

Your Turn: A Project Seed

- ▣ **Question:** does the SPY – RSP spread survive after controlling for sector returns?
 - ▶ regress d_t on the sector ETFs (XLK, XLC, XLF, ...) from the course market data; test H_0 : intercept = 0 with HAC standard errors
- ▣ **Replicate first:** SPY $\times 12.7$, RSP $\times 11.9$ and the mini-case t -statistics
- ▣ **Pre-register:** H_0 , H_1 , test, sample and corrections, committed before the results
- ▣ **Report:** estimate, standard error, MDE and every specification tried; the code as a Quantlet
- ▣ **Log:** prompts that shaped decisions in AI_USE.md, every caught error in AI_ERRORS.md; reusable prompts for each step are in the [▶ Appendix](#)

Key Takeaways

- ▣ Back to the guiding question: a comparison is trustworthy only with the right data, the right calendar and an honest interval
- ▣ Markets in 2026 combine exchanges, OTC dealers, ETFs, algorithms and blockchains; the observation time of a “daily price” depends on the venue
- ▣ Each wave of models, from random walks to machine learning and quantum computing, solved a new problem with new data and computing power; bubbles and crashes have recurred for four centuries
- ▣ Every number is an estimate
 - ▶ standard error of an annualised mean return: $\sigma_a/\sqrt{Y}$; of an annualised Sharpe ratio: about $1/\sqrt{Y}$
 - ▶ an interval that contains zero means “inconclusive”, not “equal”
- ▣ Cap-weighted passive investing carries market concentration; crypto moved closer to mainstream finance through ETFs and stablecoins
- ▣ A claim, human or AI-generated, needs an estimate, an interval, a rival explanation and a stated limitation

Check Yourself, and Next: Chapter 1

Retrieval questions

- Why does the SE of a mean return not fall when you use daily instead of monthly data?
- Why must prices be joined on common days before computing Bitcoin – S&P 500 returns?
- A Sharpe-ratio difference has a 95% interval $[-0.65, +0.81]$: what may you conclude?
- Why is the SPY/RSP ratio not a measure of passive ownership?

Chapter 1: stylised facts

- How heavy are the tails of daily returns, and is the kurtosis even finite?
- Does volatility have long memory?
- At which horizon do returns approach the Normal distribution?
- Do the facts hold for the S&P 500, Bitcoin, BVB stocks and EUR/RON?

Appendix: Course Map, 20 Chapters

- ▣ **0** Financial Markets in 2026
- ▣ **1** Stylised Facts and Returns
- ▣ **2** Market Efficiency: From EMH to Adaptive Markets
- ▣ **3** Factor Models and Asset Pricing
- ▣ **4** Portfolio Optimisation
- ▣ **5** Conditional Volatility: GARCH Models
- ▣ **6** Multivariate Volatility and Dependence
- ▣ **7** Value-at-Risk and Expected Shortfall
- ▣ **8** Backtesting and Evaluating Risk Forecasts
- ▣ **9** Realised Volatility
- ▣ **10** Market Microstructure
- ▣ **11** Continuous-Time Models
- ▣ **12** Options and the Volatility Surface
- ▣ **13** Machine Learning in Finance
- ▣ **14** Deep Learning and Time-Series Foundation Models
- ▣ **15** LLMs and Sentiment Analysis
- ▣ **16** Digital Assets and DeFi
- ▣ **17** Bubbles and Crashes
- ▣ **18** Systemic Risk and Networks
- ▣ **19** Review and Project Presentations



Appendix: Core Reading and Further Study

- ▣ Franke, Härdle & Hafner (2019), *Statistics of Financial Markets*, 5th ed.: the main statistical reference
- ▣ Campbell, Lo & MacKinlay (1997), *The Econometrics of Financial Markets*; Brandimarte (2018), *An Introduction to Financial Markets: A Quantitative Approach*
- ▣ Härdle & Simar (2019), *Applied Multivariate Statistical Analysis*; Härdle, Chen & Overbeck (2017) (Eds.), *Applied Quantitative Finance*, 3rd ed.
- ▣ Härdle, Harvey & Reule (2020), "Understanding cryptocurrencies", *Journal of Financial Econometrics*
- ▣ Advanced courses on quantinar.com
 - ▶  Statistics of Financial Markets;  Digital Transformation on Finance
 - ▶  Cryptos, NFTs, Digital Assets;  Introduction to Blockchain and Cryptocurrencies

Appendix: Asset-Class Terms

- ▣ **T-bills:** Treasury bills, government debt with maturity below one year; **repo:** repurchase agreement, a short-term loan backed by securities
- ▣ **Futures:** a contract to buy or sell later at a price fixed today; **roll yield:** the gain or loss from replacing an expiring contract with the next one
- ▣ **Options:** the right, not the obligation, to buy or sell at a fixed price; **swaps:** exchanges of cash-flow streams
- ▣ **Idiosyncratic risk:** specific to one company; **counterparty risk:** the other side of a contract may not pay
- ▣ **Multilateral trading facility (MTF):** a trading venue run outside a regulated exchange; **central counterparty (CCP):** becomes the buyer to every seller and the seller to every buyer; **custodian:** keeps the securities

Appendix: Where Trading Happens

Venue	How trades are matched	What is visible	Examples
Exchange	central limit order book, auctions	quotes and trades	NYSE, Nasdaq, BVB
OTC dealer	bilateral, with a dealer	little, after the trade	bonds, FX, swaps
Dark pool	at a price taken from public markets	nothing before the trade	large equity orders
Centralised crypto exchange	order book run by a company, 24/7	quotes and trades	spot Bitcoin, Ether
AMM (decentralised)	a pool of two tokens, price set by a formula	the pool on the blockchain	Chapter 16

- **OTC**: over the counter, negotiated between two parties without an exchange; **AMM**: automated market maker, a smart contract that trades without an operator
- Crypto trades every day of the year, on hundreds of venues: [Makarov & Schoar \(2020\)](#) find large, persistent cross-exchange price gaps that capital controls and costs keep open

Appendix: Auctions versus Continuous Trading

Call auctions

- ▣ Orders are collected, then matched at the one price that maximises traded volume
- ▣ Used at the open and the close of most exchanges
- ▣ Concentrate liquidity: the closing auction sets the daily close used in this course

Continuous trading

- ▣ Each incoming order is matched at once if possible: best price first, then the earliest order
- ▣ Speed is rewarded: **high-frequency trading (HFT)** programs trade in microseconds
- ▣ Budish, Cramton & Shim (2015) propose frequent batch auctions to end the speed race

Appendix: From the Trading Floor to Screens



NYSE trading floor, 2011

Photo: Carol M. Highsmith (2011);
public domain; Wikimedia Commons



Electronic displays, Chicago Board of Trade, 1992

Photo: Lars Plougmann (1992); CC BY-SA 2.0;
Wikimedia Commons

- **Open outcry:** traders met on the floor and matched orders by voice and hand signals
- Electronic limit order books gradually replaced most floor trading
- Today most volume is matched by computers (market microstructure, [Chapter 10](#))

Appendix: The Roll Model of Bid–Ask Bounce

- Model: $p_t = m_t + (s/2) q_t$, with m_t the efficient price (a random walk), s a constant spread in price units, $q_t = \pm 1$ independent, equally likely trade signs, unrelated to Δm_t
- Then $\Delta p_t = \Delta m_t + (s/2)(q_t - q_{t-1})$ and only the shared q_{t-1} links consecutive changes:

$$\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -\frac{s^2}{4} \text{Var}(q_{t-1}) = -\frac{s^2}{4}, \quad \hat{s} = 2\sqrt{-\hat{\gamma}_1}$$

- If $\hat{\gamma}_1 > 0$ the estimator is undefined ($\hat{\gamma}_1 = 0$ gives $\hat{s} = 0$); other dependence also moves $\hat{\gamma}_1$, so not every negative autocovariance is bounce ([Roll \(1984\)](#); [Chapter 10](#))

Appendix: A Historic Crash, Black Tuesday, 1929



Crowd outside the NYSE, 29 October 1929

Photo: U.S. government (1929); public domain;
[Wikimedia Commons](#)

- 29 October 1929 (**Black Tuesday**): crowds gather outside the New York Stock Exchange during the crash
- The crash of October 1929 deepened a US recession that had begun in August 1929; the initial contraction of the Great Depression lasted until 1933 ([NBER business-cycle dates](#))
- Long, deep drawdowns are not new; bubbles and crashes are modelled in [Chapter 17](#)

Appendix: Tulips and the South Sea Bubble, Revisited



Jan Brueghel the Younger, *Satire on Tulip Mania*, c. 1640

Painting: Jan Brueghel the Younger; public domain; [Wikimedia Commons](#)

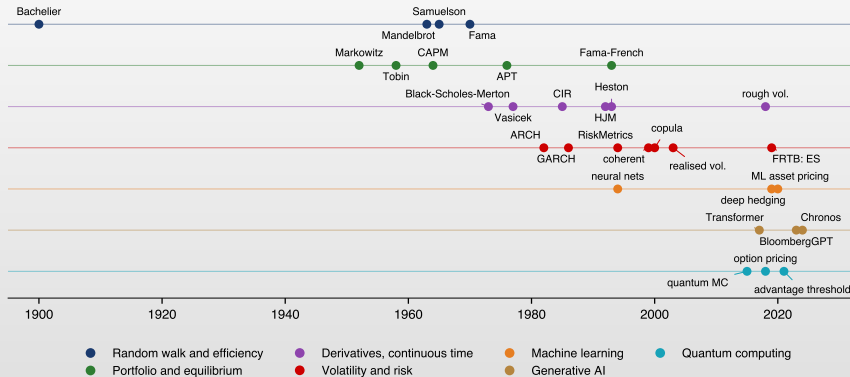


William Hogarth, *The South Sea Scheme*, 1721

Print: William Hogarth (1721); CC0; [National Gallery of Art, Wikimedia Commons](#)

- The popular story of tulip mania comes largely from moralising pamphlets written after the crash
- [Garber \(1989\)](#): prices of rare bulbs fell the way prices of new varieties fall in later flower markets; the spike in common bulbs lasted a few weeks of winter trading in taverns, with contracts that were rarely enforced
- [Goldgar \(2007\)](#): archival records show few bankruptcies and little damage to the Dutch economy
- [Frehen, Goetzmann & Rouwenhorst \(2013\)](#): in 1720 the boom spread across Paris, London and Amsterdam; prices rose most for firms tied to Atlantic trade and to insurance
- Lesson: a bubble needs a story about a real change; dating it requires a model of fundamental value ([Chapter 17](#))

Appendix: Financial Models, 1900–2026: an Overview



Appendix: Origins: Random Walks and Efficient Markets



Louis Bachelier

Photo: unknown author; public domain; Wikimedia Commons

- [Bachelier \(1900\)](#), *Théorie de la spéculation*: prices of French government bonds follow a Brownian motion, five years before Einstein; the first option-pricing formula
- [Mandelbrot \(1963\)](#): cotton price changes have far heavier tails than the Normal distribution allows
- [Samuelson \(1965\)](#): geometric Brownian motion keeps prices positive and makes log returns Normal
- [Fama \(1965\)](#); [Fama \(1970\)](#): the EMH (efficient market hypothesis), prices fully reflect available information
 - ▶ weak form (past prices), semi-strong form (public information), strong form (all information)
- In this course: stylised facts ([Chapter 1](#)), efficiency tests ([Chapter 2](#)), continuous-time models ([Chapter 11](#))

Appendix: Portfolios and Equilibrium Pricing



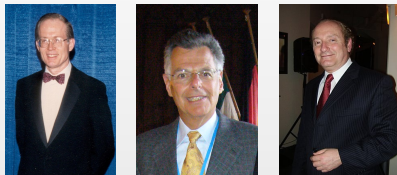
William F. Sharpe, 2007

Photo: Larry D. Moore (2007); CC

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- [Markowitz \(1952\)](#): choose portfolio weights by mean and variance; diversification depends on covariances
- [Tobin \(1958\)](#): with a riskless asset, every investor holds the same risky portfolio (two-fund separation)
- CAPM (capital asset pricing model): $\mathbb{E}[R_i] - r_f = \beta_i (\mathbb{E}[R_m] - r_f)$, only market risk is priced ([Sharpe \(1964\)](#); [Lintner \(1965\)](#); [Mossin \(1966\)](#))
 - ▶ R_i, R_m : returns of asset i and of the market portfolio; r_f : risk-free rate;
 $\beta_i = \text{Cov}(R_i, R_m) / \text{Var}(R_m)$
 - ▶ assumptions: mean–variance investors with common beliefs, borrowing and lending at r_f , no frictions
- APT (arbitrage pricing theory): no arbitrage with K common factors gives a linear factor model ([Ross \(1976\)](#))
- Three factors: market, size (SMB, small minus big) and value (HML, high minus low book-to-market) ([Fama & French \(1993\)](#))
- In this course: factor models ([Chapter 3](#)), portfolio optimisation ([Chapter 4](#))

Appendix: Derivatives and Continuous Time



Fischer Black, Myron Scholes, Robert C. Merton

Black: Dalmatine; public domain; Scholes:

Nobellaureatesphotographer (2008); CC BY-SA 3.0;

Merton: Digarnick (2006); public domain; Wikimedia

Commons

- Black & Scholes (1973); Merton (1973): a continuously rebalanced hedge replicates an option, so its price does not depend on the expected return of the stock
- Heston (1993): stochastic volatility with a closed-form option price
- Gatheral, Jaisson & Rosenbaum (2018): log-volatility behaves like a fractional Brownian motion with a Hurst exponent near 0.1 (rough volatility)
 - ▶ fractional Brownian motion B^H : Gaussian, with $\text{Var}(B_{t+h}^H - B_t^H) = |h|^{2H}$; $H = 1/2$ is Brownian motion, $H < 1/2$ gives rougher paths
- In this course: Chapters 11 and 12

Appendix: Interest-Rate Models

□ Interest rates

- ▶ Vasicek (1977): $dr_t = \kappa(\theta - r_t) dt + \sigma dW_t$, the short rate reverts to the level θ but can become negative
- ▶ CIR (Cox, Ingersoll & Ross (1985)): $dr_t = \kappa(\theta - r_t) dt + \sigma\sqrt{r_t} dW_t$; the rate's volatility shrinks as the rate approaches zero, so the rate does not become negative
- ▶ HJM (Heath, Jarrow & Morton (1992)): models the whole forward-rate curve directly; the no-arbitrage condition fixes each rate's drift as a function of its volatilities

Appendix: Volatility Models



Robert F. Engle, 2022

Photo: Jon Demske (2022); CC

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- r_t : daily return (demeaned), σ_t^2 : its variance conditional on the past
- ARCH (autoregressive conditional heteroskedasticity, [Engle \(1982\)](#)): $\sigma_t^2 = \omega + \alpha r_{t-1}^2$, today's variance depends on yesterday's squared shock
- GARCH (generalised ARCH, [Bollerslev \(1986\)](#)): $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$ adds the lagged variance
- RiskMetrics (J.P. Morgan, 1994): an EWMA (exponentially weighted moving average) of squared returns, $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$, with $\lambda = 0.94$ for daily data ([J.P. Morgan/Reuters \(1996\)](#))
 - ▶ the return k days back gets weight $(1 - \lambda)\lambda^{k-1}$: a GARCH(1,1) with $\omega = 0$ and $\alpha + \beta = 1$
- Realised volatility from intraday returns ([Andersen et al. \(2003\)](#))
- In this course: [Chapters 5–9](#)

Appendix: Risk Measurement and Regulation

- ▣ VaR (value at risk) of a return X at tail probability α : $\text{VaR}_\alpha(X) = -q_\alpha(X)$, the loss exceeded with probability α ; q_α is the α -quantile and “VaR 1%” means $\alpha = 1\%$
- ▣ ES (expected shortfall): $\text{ES}_\alpha(X) = -\frac{1}{\alpha} \int_0^\alpha q_u(X) du$, the mean loss in the worst α of cases ($= -\mathbb{E}[X \mid X \leq q_\alpha(X)]$ for continuous distributions)
- ▣ 1996: the BCBS (Basel Committee on Banking Supervision) lets banks compute market-risk capital with internal models of the 10-day VaR 1% (BCBS (1996)); the FRTB (fundamental review of the trading book) replaces it with ES 2.5% (BCBS (2019))
- ▣ **Coherent** risk measure (Artzner et al. (1999)): monotone, translation invariant, positively homogeneous and subadditive, $\rho(X + Y) \leq \rho(X) + \rho(Y)$
 - ▶ VaR can violate subadditivity and so penalise diversification; ES is coherent (Acerbi & Tasche (2002))
- ▣ EVT (extreme value theory) models the tails beyond most of the data (Embrechts, Klüppelberg & Mikosch (1997); McNeil & Frey (2000))
- ▣ How the Gaussian copula mispriced the tranches of CDOs (collateralised debt obligations) before 2008: next slide

Appendix: The Gaussian Copula and 2008

- [Li \(2000\)](#): joint default times from marginal default probabilities F_i and a Normal dependence structure

$$\tau_i = F_i^{-1}(\Phi(Z_i)), \quad Z_i = \sqrt{\rho} M + \sqrt{1 - \rho} \varepsilon_i, \quad M, \varepsilon_i \sim \mathcal{N}(0, 1) \text{ independent}, \quad 0 \leq \rho < 1$$

- One parameter ρ priced every tranche of a CDO, a security that pays from a pool of loans in order of seniority
- Drawbacks discussed after 2007–2008 ([Donnelly & Embrechts \(2010\)](#))
 - ▶ zero asymptotic tail dependence: far enough in the tails, defaults become independent, so default clustering in a crisis is modelled poorly; this limit alone does not prove mispricing at the default thresholds actually priced
 - ▶ the correlation implied by market prices differed across tranches, so one ρ could not price them all consistently
 - ▶ no economic factors in the model, so stress tests are hard to build
- Lesson: model risk; dependence in the tails ([Chapter 6](#)) and systemic risk ([Chapter 18](#))

Appendix: Machine Learning in Finance

- ML (machine learning): flexible functions fitted to data and selected by out-of-sample performance rather than by theory
- [Hutchinson, Lo & Poggio \(1994\)](#): neural networks learn option prices and hedge ratios from data, without the Black–Scholes formula
- [Gu, Kelly & Xiu \(2020\)](#): about 30,000 US stocks, 1957–2016, 94 firm characteristics
 - ▶ trees and neural networks beat linear models out of sample, through nonlinear interactions
 - ▶ out-of-sample $R_{oos}^2 = 1 - \sum (r_{i,t+1} - \hat{r}_{i,t+1})^2 / \sum r_{i,t+1}^2$ over the test months: the benchmark forecasts a zero excess return, so the denominator is not demeaned
 - ▶ the monthly R_{oos}^2 stays below 1%: a small but economically valuable signal
- [Buehler et al. \(2019\)](#), deep hedging: a neural network chooses hedge positions under transaction costs to minimise a convex risk measure, with no pricing model
- Main risks: overfitting, look-ahead and many trials on the same data ([Chapters 13 and 14](#))

Appendix: Generative AI and Foundation Models

- ▣ The Transformer ([Vaswani et al. \(2017\)](#)): attention layers read a whole sequence in parallel; it underlies most current LLMs (large language models) and GPT (generative pre-trained Transformer)
- ▣ BloombergGPT ([Wu et al. \(2023\)](#)): a 50-billion-parameter LLM; its corpus held 363 billion tokens of financial text and 345 billion general tokens, of which 569 billion were used in training
- ▣ [Lopez-Lira & Tang \(2026\)](#): ChatGPT scores of news headlines predict next-day stock returns out of sample
- ▣ Foundation models for time series: Chronos scales and quantises values into tokens and trains a language model on many series, then forecasts new series zero-shot ([Ansari et al. \(2024\)](#))
- ▣ Open problems
 - ▶ look-ahead: the model may have read texts written after the forecast date
 - ▶ cost, reproducibility and the decay of a signal once many traders use it
- ▣ In this course: [Chapters 14 and 15](#)

Appendix: Quantum Computing: Promise and Limits



IBM Quantum System One, Ehningen
(Germany), 2021

Photo: IBM Research (2021); CC BY 2.0;
Wikimedia Commons

- Qubits hold superpositions, and interference amplifies the right answer (Orús, Muga & Lizaso (2019); Herman et al. (2023))
- **Proven:** to estimate a mean, e.g. an option price as a discounted expected payoff, with error ε , classical MC (Monte Carlo) needs about $1/\varepsilon^2$ samples, QAE (quantum amplitude estimation) about $1/\varepsilon$ steps (Montanaro (2015))
 - ▶ $\varepsilon = 10^{-4}$: 10^8 samples against 10^4 steps; the gain is *quadratic*: the number of steps becomes the square root of the classical one, it does not fall exponentially
 - ▶ a VaR is a quantile, not a mean: QAE estimates $P(X \leq x)$ and a bisection search over x finds q_α , at an extra cost of a few QAE runs (Woerner & Egger (2019))
- **Demonstrated** on a few qubits: option pricing (Rebentrost, Gupta & Bromley (2018); Stamatopoulos et al. (2020)), VaR and ES (Woerner & Egger (2019))
- The calculation step by step: next slide

Appendix: Classical Monte Carlo versus Quantum Amplitude Estimation

- ▣ Classical MC: $\hat{\mu}_N = \frac{1}{N} \sum_{j=1}^N f(X_j)$ has standard error $\sigma/\sqrt{N}$, so accuracy ε needs $N = O(\sigma^2/\varepsilon^2)$ samples
- ▣ QAE prepares $\sqrt{1-\mu}|\psi_0\rangle|0\rangle + \sqrt{\mu}|\psi_1\rangle|1\rangle$ (for f scaled to $[0, 1]$): μ is the probability of measuring 1 on the objective qubit, entangled with the register that holds X (Woerner & Egger (2019))
- ▣ A Grover-type operator applied m times makes the error fall as $O(1/m)$: $O(1/\varepsilon)$ calls of the pricing circuit (Montanaro (2015); Rebentrost, Gupt & Bromley (2018))
- ▣ *Quadratic* means: where classical MC takes N steps, QAE takes about $\sqrt{N}$; an exponential gain would mean about $\log N$ steps, but in the oracle-query model no quantum algorithm estimates a mean with fewer calls in the worst case, up to logarithmic factors (Montanaro (2015))
 - ▶ $\varepsilon = 10^{-4}$: about 10^8 classical samples against about 10^4 quantum calls
- ▣ Why this is not yet an advantage
 - ▶ each call must load the distribution of X and run without error, which needs error correction
 - ▶ error-corrected operations are far slower than classical ones: for their autocallable and TARF benchmarks, Chakrabarti et al. (2021) estimate about 8 000 logical qubits and a T-depth (sequential layers of the slowest error-corrected gate) of 54 million
 - ▶ classical MC parallelises easily across many processors

Appendix: Quantum Computing: Why It Is Not Yet Practical

- ▣ [Preskill \(2018\)](#) called the hardware of 2018 NISQ (Noisy Intermediate-Scale Quantum): tens to a few hundred noisy qubits without error correction
- ▣ Physical qubit counts have grown since: IBM's Condor chip had 1 121 qubits in 2023, and error mitigation extracted useful output from circuits with more than 100 qubits and 2 000 operations ([IBM Research \(2024\)](#))
- ▣ What limits applications is not the physical count
 - ▶ **logical qubits**: error-corrected qubits, each built from many physical ones
 - ▶ usable circuit depth before noise dominates, set by error rates, connectivity, error mitigation and error correction
 - ▶ loading a return distribution into qubits is itself costly
- ▣ For their benchmark derivatives, autocallables (notes redeemed early when the underlying crosses a barrier) and a TARF (target accrual redemption forward), [Chakrabarti et al. \(2021\)](#) estimate about 8 000 logical qubits and a T-depth of 54 million, run in about a second
 - ▶ an upper bound for these products and algorithms, not a universal threshold; still far beyond current machines

Appendix: What Changed Each Time, and Where It Appears in This Course

Wave	What made it possible	Chapters
Random walk, efficiency	probability theory, long price series	1, 2, 11
Portfolio, equilibrium	utility theory, the first stock-return databases	3, 4
Derivatives, continuous time	stochastic calculus, listed options (1973)	11, 12
Volatility and risk	daily and intraday data, bank regulation	5–9
Machine learning	large panels, cheap computing	13
Generative AI	text at scale, GPU (graphics processing unit)	14, 15
	hardware, the Transformer	
Quantum computing	new hardware, not yet practical	outlook

- No wave replaced the earlier ones: GARCH still feeds VaR, factor models still benchmark ML, Black–Scholes still quotes implied volatility
- The complete course map is the first slide of this Appendix

Appendix: Risk, Dependence and Rates, Notation

- ▣ **Drawdown** $DD_t = P_t / \max_{s \leq t} P_s - 1$; **maximum drawdown (MDD)** $= \min_t DD_t$, the worst loss from a previous peak
- ▣ **Correlation**: Pearson correlation of returns, never of prices; **rolling**: on a moving 252-day window
 - ▶ two calendars (stocks vs crypto): join the *prices* on common days first, then compute returns
- ▣ **Yield to maturity**: the internal rate of return of a bond's promised cash flows
 - ▶ equals the realised return only if the bond is held to maturity, pays in full and coupons are reinvested at the same yield
- ▣ **Yield-curve spread**: the 10-year government bond yield minus the 2-year yield, $y^{10y} - y^{2y}$
 - ▶ measured in percentage points (pp): $4.3\% - 3.8\% = 0.5$ pp; markets also quote basis points (bp), $1 \text{ bp} = 0.01 \text{ pp}$, so $0.5 \text{ pp} = 50 \text{ bp}$
 - ▶ **inverted curve**: a negative spread, i.e. the 2-year rate exceeds the 10-year rate; in the US it has usually preceded recessions (Estrella & Mishkin (1998))

Appendix: $\sqrt{h}$ Scaling and the Variance Ratio

- Stationary returns with variance σ^2 and autocorrelations ρ_j :

$$\text{Var}\left(\sum_{j=1}^h r_{t+j}\right) = h\sigma^2 \left[1 + 2 \sum_{j=1}^{h-1} \left(1 - \frac{j}{h}\right) \rho_j\right]$$

- The $\sqrt{h}$ rule is exact when returns are **uncorrelated** with **constant variance**; independence is not needed
- Variance ratio** $\text{VR}(h) = \text{Var}(h\text{-day return})/(h\sigma^2)$; for an AR(1), $\text{VR}(h) \rightarrow (1 + \rho)/(1 - \rho)$
- S&P 500, 2000–2026: $\hat{\rho}_1 = -0.10$ and $\widehat{\text{VR}}(20) = 0.75$
 - scaling daily volatility by $\sqrt{20}$ overstates the 20-day (monthly) risk
 - use the whole autocorrelation function or aggregate the returns directly (more examples on the next slide; tests in [Chapter 2](#))

Appendix: Variance Ratios and Variance Estimates, More Examples

- EUR/RON, BNR reference rate, 2005–2026: $\hat{\rho}_1 = 0.17$
 - ▶ a hypothetical AR(1) would give the factor $\sqrt{1.17/0.83} = 1.19$ (the 2005–2026 volatility annualised with $\sqrt{252}$, 4.4%, would become 5.3%)
 - ▶ but $\hat{\rho}_2, \hat{\rho}_3, \hat{\rho}_4 < 0$ offset lag 1: monthly returns give 4.4% (Seminar 0, A6)
- EUR/RON series from EODHD: bad ticks give $\hat{\rho}_1 = -0.57$ and inflate $\sqrt{252}$ volatility; they are errors to remove, not an AR(1)
- Variance behaves differently from the mean: for iid returns with a finite fourth moment, $\text{Var}(\hat{\sigma}^2) \approx \sigma^4(\kappa - 1)/T$, with kurtosis $\kappa = \mathbb{E}[(r_t - \mu)^4]/\sigma^4$; it shrinks slowly when κ is large (SPY daily: $\kappa = 17.1$, [Chapter 1](#))
 - ▶ dependent returns (volatility clustering): $\sigma^4(\kappa - 1)$ becomes the long-run variance of $(r_t - \mu)^2$, which adds the autocovariances of squared returns; with an infinite fourth moment the approximation fails

Appendix: The Sharpe Ratio, Delta Method, Fat Tails, Dependence and Selection

- Delta method, iid returns following the Normal distribution (Lo (2002)):

$$\sqrt{T}(\widehat{\text{SR}}_d - \text{SR}_d) \xrightarrow{d} N(0, 1 + \text{SR}_d^2/2) \Rightarrow \text{SE}(\widehat{\text{SR}}_a) = \sqrt{\frac{1 + \text{SR}_d^2/(2q)}{Y}} \approx \frac{1}{\sqrt{Y}}$$

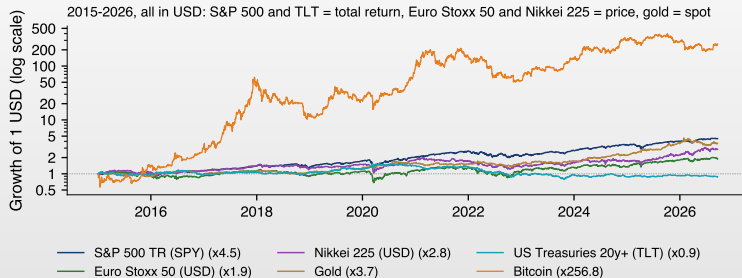
- Non-Normal iid returns with skewness γ_3 and kurtosis γ_4 : the asymptotic variance of $\sqrt{T}(\widehat{\text{SR}}_d - \text{SR}_d)$ is

$$V = 1 - \text{SR}_d\gamma_3 + \text{SR}_d^2(\gamma_4 - 1)/4, \quad \text{SE}(\widehat{\text{SR}}_a) = \sqrt{qV/T}$$

► Opdyke (2007); SPY: SE 0.30 instead of 0.29, because the daily SR is small

- Dependent returns: HAC or studentised bootstrap inference (Ledoit & Wolf (2008))
- Selection: picking the best of many assets or strategies ex post inflates the SR; the deflated SR corrects for it (Bailey & López de Prado (2014), Chapter 13)

Appendix: Asset Classes since 2015

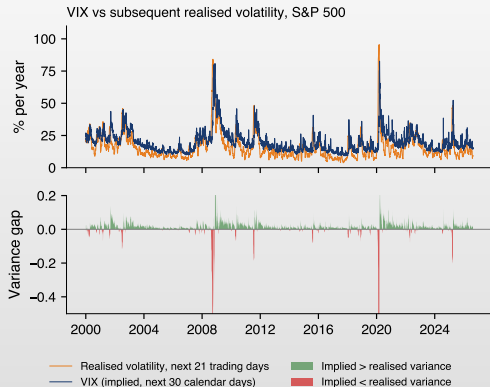


- Value of 1 USD invested on 2 Jan 2015, log scale; Euro Stoxx 50 and Nikkei 225 converted with daily [FRED](#) exchange rates
- SPY and TLT (iShares 20+ Year Treasury Bond ETF) are total return; Euro Stoxx 50 and Nikkei 225 are price indices, without dividends
- Long Treasuries ended below their 2015 level ($\times 0.9$); Bitcoin's $\times 257$ is one coin chosen ex post among thousands

Appendix: Selection, Survivorship and Horizon Effects

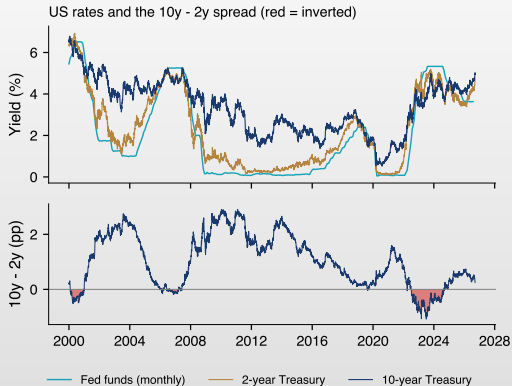
- ▣ **Survivorship:** samples rebuilt from surviving firms and coins make past returns look too good (Brown, Goetzmann, Ibbotson & Ross (1992))
- ▣ **Selection:** Bitcoin's $\times 257$ is chosen ex post among thousands of coins; a best-of-many Sharpe ratio needs a deflated SR
- ▣ **Horizon:** the depth of the maximum drawdown grows with the sample length
 - ▶ log price a driftless Brownian motion: the log drawdown $D = \max_{s \leq t \leq T} (\ln P_s - \ln P_t) \geq 0$ has $E[D] = \sqrt{\pi/2} \sigma \sqrt{T}$ (Magdon-Ismail et al. (2004))
 - ▶ $\sigma = 17.6\%$: $E[D] = 0.70$ over 10 years, 1.40 over 40; as price declines $e^{-E[D]} - 1 = -50\%$ and -75%
 - ▶ the expected percentage MDD, $E[e^{-D}] - 1$, is smaller in magnitude (Jensen): about -47% and -71% in simulated paths
- ▣ The BET's -82.5% (29 years) and an MDD from 11.7 years of data are not comparable without the same T

Appendix: Implied versus Realised Variance



- $(\text{VIX}/100)^2 \approx$ the **risk-neutral** expectation of the annualised S&P 500 variance over the next 30 days
- **Variance risk premium:** risk-neutral minus real-world expected variance, positive on average ([Carr & Wu \(2009\)](#))
- Proxy here: realised volatility over the next 21 trading days, $\sqrt{(252/21) \sum r^2}$; incomplete final windows omitted
- 2000–2026: mean VIX 19.8 vs realised 16.2; the VIX was higher on 83% of days
- Ex post the gap is premium plus forecast error; [Bollerslev, Tauchen & Zhou \(2009\)](#) show that a proxy for the variance risk premium, known at t , predicts quarterly equity returns

Appendix: Interest Rates and the Yield Curve



- **FRED**: 10-year and 2-year US Treasury yields; **fed funds rate**, the overnight rate targeted by the US central bank
- Spread 10y - 2y: briefly negative on 1–4 Apr 2022, then inverted from 6 Jul 2022 to 26 Aug 2024 (537 trading days, the longest spell), with isolated negative days on 3 and 5 Sep 2024; minimum -1.08 pp (3 Jul 2023)
- 18 Sep 2026: 10-year 5.01%, 2-year 4.76%, spread $+0.25$ pp = 25 bp; fed funds 3.63% (Aug 2026)
- A yield is not a realised return: TLT lost value in 2015–2026 as yields rose

Appendix: Two Traps for Crypto–Equity Correlations

- **Asynchronous closes:** Bitcoin's day ends at 24:00 UTC, the S&P 500's 3–4 hours earlier, so part of the co-movement appears at lags
 - ▶ Bitcoin – S&P 500, 2022–2026: same day 0.42; S&P 500 with Bitcoin one day earlier +0.03, one day later –0.05; sum 0.40
 - ▶ diagnostic: the asynchronous-close bias is small for this pair at the daily frequency
- **Volatility bias:** correlations estimated on turbulent days are larger even when the link is stable ([Forbes & Rigobon \(2002\)](#))
 - ▶ VIX in its top quartile (> 21.8): 0.52 vs 0.32 on the other days
 - ▶ adjusted for the 4.64 times higher S&P 500 variance: 0.27, so the “stress correlation” is mostly a volatility effect
- Models, assumptions and correction formulas: [▶ asynchronous closes](#) and [▶ volatility bias](#)

 [MFM_ch0_correlations](#)

Appendix: Asynchronous Closes

- Bitcoin's price for day t is taken at 24:00 UTC; the S&P 500 closes at 20:00 or 21:00 UTC: news arriving between the two times enters the two series on different days
- With a common factor, the same-day correlation is attenuated and the rest appears in lead/lag correlations
 - ▶ if only the common factor is split across days and daily variances are unchanged, the lead, same-day and lag correlations sum to the true one (Scholes & Williams (1977); Lo & MacKinlay (1990); Seminar 0, A9)
 - ▶ otherwise use longer returns (Burns, Engle & Mezrich (1998))
- Bitcoin – S&P 500, 2022–2026, common days: $0.42 + 0.03 - 0.05 = 0.40$; the sum is a diagnostic, not a proof

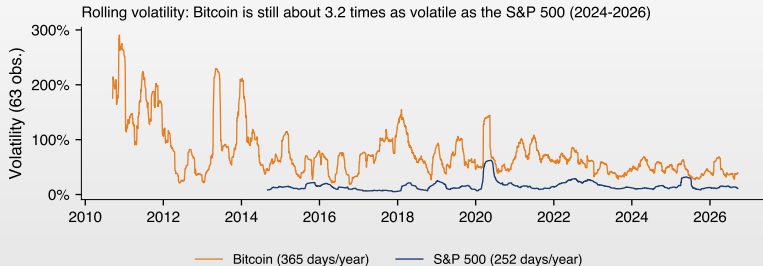
Appendix: Correlation in Turbulent Periods, the Volatility Bias

- Stable link $y = \beta x + \varepsilon$ (β and $\text{Var} \varepsilon$ fixed) with correlation ρ ; estimated only on days when $\text{Var}(x)$, the source market, is $(1 + \delta)$ times higher, it gives ([Forbes & Rigobon \(2002\)](#))

$$\rho^* = \rho \sqrt{\frac{1 + \delta}{1 + \delta \rho^2}}, \quad |\rho^*| > |\rho| \text{ for } \delta > 0, \quad 0 < |\rho| < 1$$

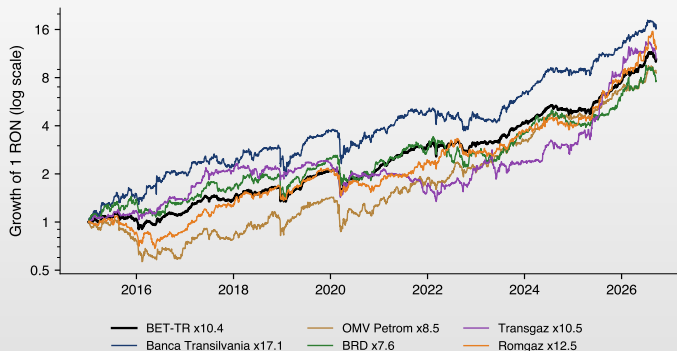
- Bitcoin – S&P 500, 2022–2026, days split by the VIX (top quartile: $\text{VIX} > 21.8$)
 - ▶ 0.52 on high-VIX days vs 0.32 on the others; S&P 500 variance ratio $1 + \delta = 4.64$
 - ▶ adjusted $\hat{\rho} = \rho^* / \sqrt{1 + \delta(1 - \rho^{*2})} = 0.27$
- The adjustment assumes no omitted common shock and no feedback between the two markets

Appendix: Bitcoin Remains Several Times as Volatile as Equities



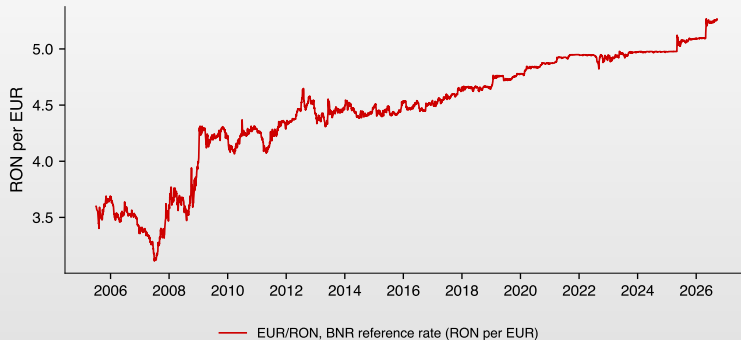
- Rolling annualised volatility over the last 63 observations: about 3 months for the S&P 500 (annualised with $\sqrt{252}$), about 2 months for Bitcoin (annualised with $\sqrt{365}$)
- Bitcoin: 65% on average in 2015–2017, 46% in 2024–2026; S&P 500 in 2024–2026: 14%, so Bitcoin is still about **3.2 times** as volatile
- Volatility changes over time for both: a constant-volatility model is a poor description

Appendix: Romanian Equities since 2015



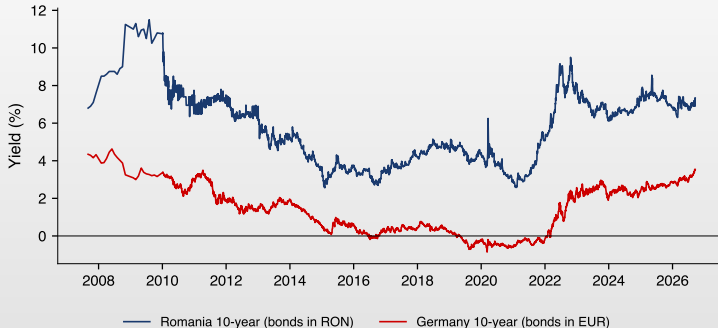
- Growth of 1 RON invested in January 2015, dividend-adjusted (adjusted close), log scale
- BET-TR $\times 10.4$, Banca Transilvania $\times 17.1$: over this window the large banks and energy firms carried the index

Appendix: The Leu, the EUR/RON Reference Rate



- EUR/RON (BNR reference rate) from 3.60 (4 Jul 2005, after the redenomination 1 RON = 10 000 ROL) to 5.26
- A managed currency: annualised volatility of 2.1% in 2015–2026, long calm spells and occasional steps

Appendix: Romanian and German 10-Year Government Yields



- 18 Sep 2026: Romania 7.33% (bonds in RON), Germany 3.53% (in EUR): a cross-currency differential of 3.8 pp
- The differential mixes expected leu depreciation, inflation and policy-rate gaps with credit, liquidity and term premia: it is **not** a pure sovereign-credit spread

Appendix: The Bucharest Stock Exchange, 1882–1948

- **1 Dec 1882:** the securities and commodities exchange opens under a royal decree of King Carol I [BVB \(history\)](#)
- Early shocks [BVB press release \(2017\)](#)
 - ▶ the 1907 peasants' revolt and the Balkan wars of 1912–1913
 - ▶ closed during the First World War
- After reopening: about 7 years of strong growth, then about 7 years of accelerated decline, overlapping the Great Depression of 1929–1933 [BVB press release \(2017\)](#)
- **1929:** a new law on exchanges, with the explanatory memorandum of minister Virgil Madgearu [Lege asupra burselor \(1929\)](#)
- **1948:** trading stops with the communist regime and the nationalisation of the economy [BVB \(history\)](#)



Share of UDR (Uzinele și Domeniile Reșița), June

1926. Scan: EDHAC e.V.; public domain;

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Appendix: Palatul Bursei, the Exchange's Historic Home



Photo: Nicolae Ionescu (1928); public domain; Wikimedia Commons



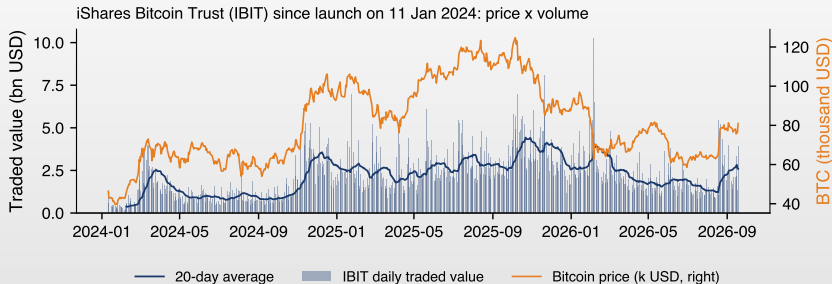
Photo: Chainwit. (2023); CC BY 4.0; Wikimedia Commons

- Built 1906–1912, architect Ștefan Burcuș, Str. Ion Ghica 4 [Palatul Bursei](#)
- Seat of the Exchange and of the Bucharest Chamber of Commerce and Industry
- From 1955: the State Central Library (today the National Library)
- Left: stockbrokers at the entrance, March 1928; right: the building in 2023

Appendix: Interruption and Rebirth, 1948–2023

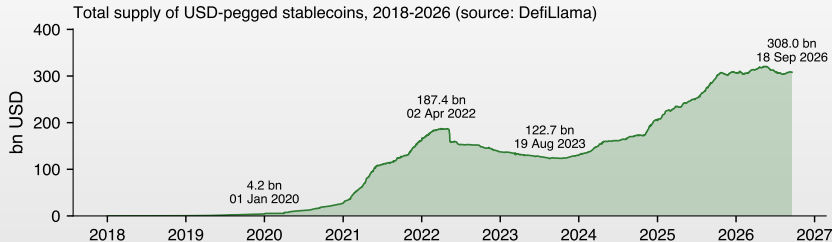
- ▣ **1948–1995:** almost 50 years without a stock exchange [BVB press release \(2017\)](#)
- ▣ **20 Nov 1995:** first session of the re-established exchange: 905 shares of 6 listed companies [BVB press release \(2017\)](#)
- ▣ **1997:** launch of BET on 19 Sep 1997 at 1 000 points, with 10 companies; the basket may hold 10–20 and today has 20 [BVB \(indices\)](#)
 - ▶ **BET-FI:** the first sector index, for the financial investment companies (SIFs, listed in 1999)
- ▣ **2010:** BVB lists its own shares on its **regulated market** (an EU-authorized trading venue) [BVB \(history\)](#)
- ▣ **Sep 2020:** **FTSE Russell**, a global index provider, reclassifies Romania from Frontier to (Secondary) Emerging [BVB \(history\)](#)
- ▣ **Jul 2023:** Hidroelectrica lists through the largest IPO in BVB history, EUR 1.9 bn [BVB \(history\)](#)
- ▣ Largest daily BET move: 7 Jan 2009, −12.3% (−13.1% log), after +4.8% the day before (official BVB daily archive [BVB \(indices\)](#))

Appendix: Spot Bitcoin ETFs



- **Spot ETF**: holds the asset itself, not futures; **ETP** (exchange-traded product): the wider category; spot Bitcoin ETPs approved in the US on 10 Jan 2024 ([SEC \(2024\)](#))
- IBIT (iShares Bitcoin Trust), trading since 11 Jan 2024: daily traded value = price $\times$ shares traded; median 1.97 bn USD a day, peak 10.3 bn USD (5 Feb 2026)

Appendix: Stablecoin Supply



- **Supply:** the USD value of USD-pegged stablecoins in circulation (DefiLlama): 4.2 bn (1 Jan 2020), peak 187.4 bn (2 Apr 2022), trough 122.7 bn (19 Aug 2023), 308.0 bn (18 Sep 2026)
- The 2022–2023 fall followed the TerraUSD collapse and the crypto crash; circulation value is distinct from the peg: supply can fall while the price stays at 1 USD

Appendix: Tokenisation and Regulation

Tokenised real-world assets

- ▣ **RWA** (real-world assets): bonds, fund shares, deposits, commodities
 - ▣ A token on a shared ledger is a claim on such an asset: transfer and settlement become programmable and near-instant
 - ▣ Examples: tokenised money-market and Treasury funds
 - ▣ **BIS (2023)**: a “unified ledger”; **BIS (2025)**: stablecoins fail the tests for money ([▶ details](#))
- ▣ Statistical question: do tokenised assets inherit the properties of the underlying, or those of crypto markets?

Regulation, key dates

- ▣ EU: **MiCA, Regulation (EU) 2023/1114** (Markets in Crypto-Assets)
- ▣ US: spot Bitcoin ETPs approved on 10 Jan 2024 (**SEC (2024)**); payment stablecoins under the **GENIUS Act, Public Law 119-27**
- ▣ Regulatory dates are natural break points and event-study dates

Appendix: Regulation of Digital Assets

	European Union	United States
Framework	MiCA, Regulation (EU) 2023/1114	GENIUS Act, Public Law 119-27 for payment stablecoins
Crypto ETPs	exchange-traded notes on crypto	spot Bitcoin ETPs approved in January 2024 (SEC (2024))
Supervisors	ESMA, EBA, national authorities	SEC, CFTC, banking regulators

- ESMA: European Securities and Markets Authority; EBA: European Banking Authority; SEC: Securities and Exchange Commission; CFTC: Commodity Futures Trading Commission
- BIS (Bank for International Settlements, the bank of central banks): BIS (2023) propose a “unified ledger” combining central bank money, deposits and assets; BIS (2025): stablecoins fall short of the three tests for money, **singleness** (accepted at par by all), **elasticity** (supply follows payment needs) and **integrity** (resists financial crime)

Appendix: ETF Premium Dynamics

- Statistical view of the arbitrage: an AR(1) premium

$$\pi_t = \phi \pi_{t-1} + u_t, \quad \text{half-life} = \frac{\ln 0.5}{\ln \phi}, \quad 0 < \phi < 1$$

- ϕ close to 1: premiums decay slowly, as for ETFs on foreign stocks or illiquid bonds ([Petajisto \(2017\)](#))
- In the US, in-kind redemptions also let the fund avoid distributing capital gains (a US tax rule, not a general feature)

Appendix: Case Study, How the Elasticity Is Estimated

- **Strategic response** χ : $\mathcal{E}_i = \bar{\mathcal{E}}_i - \chi \mathcal{E}_{agg}$ (eq. 4), with $\bar{\mathcal{E}}_i$ the investor's own component
 - ▶ $\chi = 0$: nobody reacts, the demand system of [Kojien & Yogo \(2019\)](#); $\chi \rightarrow \infty$: the others fully offset any change
- **Data**: quarterly holdings of US institutional managers (SEC Form 13F), 2001–2020, about 80% of US market capitalisation
- **Identification**: the same investor meets a different mix of other investors in each stock, which separates $\bar{\mathcal{E}}_i$ from χ (Theorem 1)
 - ▶ instruments: the money that would flow to each stock if the other investors held equal weights in their usual universes
- **Estimation**: GMM (generalised method of moments) with a nested equilibrium; the full demand system, equilibrium conditions and instruments are on the [next slides](#)

Appendix: Case Study, Demand System and Estimation (1/2)

- **Data:** quarterly Form 13F holdings, filed with the SEC by institutional managers with discretion over at least USD 100 million in 13(f) securities, 2001–2020; 80% of US market capitalisation (Section III.B)
 - ▶ characteristics X_k : log book equity, profitability, investment and dividend yield (dividends divided by book equity), each transformed to follow the Normal distribution, plus the square of normalised log book equity; universe $\mathcal{N}_i$: stocks held in the past three years

- **Demand** (eq. 13–14):

$$\log(w_{ik}/w_{i0}) - p_k = \bar{d}_{0i} + \bar{d}'_{1i}X_k - \mathcal{E}_{ik} p_k + \xi \mathcal{E}_{agg,k} + \zeta' \mathcal{E}_{agg,k} X_k + \epsilon_{ik}$$

- ▶ w_{ik} : weight of stock k in the portfolio of investor i , w_{i0} : the outside asset; p_k : log market capitalisation, $P_k = e^{p_k}$ (shares normalised to 1)
- ▶ $\bar{d}_{0i}$, $\bar{d}_{1i}$: demand intercept and characteristic loadings; ϵ_{ik} : demand shock (residual)
- ▶ ξ , ζ : loadings on $\mathcal{E}_{agg,k}$ and on its interaction with X_k , both from the reference price
- ▶ active: $\mathcal{E}_{ik} = \bar{\mathcal{E}}_{0i} + \bar{\mathcal{E}}'_{1i}X_k - \chi \mathcal{E}_{agg,k}$, with baseline elasticity intercept $\bar{\mathcal{E}}_{0i}$, loadings $\bar{\mathcal{E}}_{1i}$ and strategic response χ ; passive: $\mathcal{E}_{ik} = 0$
- ▶ baseline estimates: $\bar{\mathcal{E}}_{1i}$ and ζ restricted to 0 on every characteristic except book equity (Section III.B)

Appendix: Case Study, Demand System and Estimation (2/2)

- **Equilibrium** (eq. 17–18): $\sum_i w_{ik} A_i = P_k$ and $\sum_i (w_{ik} A_i / P_k) \mathcal{E}_{ik} = \mathcal{E}_{agg,k}$, A_i the investor's assets
- **Instruments**: price (eq. 23) $\hat{p}_{k,i} = \log \sum_{j \neq i} A_j 1\{k \in \mathcal{N}_j\} / |\mathcal{N}_j|$; elasticity (eq. 24): the elasticity condition solved with the same equal-weighted portfolios
- **Estimator**: GMM with a nested fixed point solved by Newton's method (Section III.C); investors with fewer than 1 000 holdings pooled by AUM (assets under management) into groups of about 2 000 holdings (Table 2)

Appendix: Case Study, Complete Results

Topic	Number in the paper	Where	Reading
Strategic response	$\chi = 2.97$ (SE 0.47); $\zeta = -0.48$ (SE 0.14)	Table 2, row 1	between 0 and ∞
Robustness	χ from 2.30 to 3.27 in rows 2–14; 0.04 without the elasticity instrument, 0.63 without instruments	Table 2	endogeneity hides the response
Elasticity level	mean $\mathcal{E}_{agg}$ 0.438 (fixed elasticities 0.39); residual cross-sectional SD 0.0395 (fixed 0.0842)	Table 3	reactions halve the dispersion
Rise of passive	median active share 81.4% (2001), 59.5% (2020)	Figure 5	passive 19% to 41%
Reduced form	slope of $\Delta \log \mathcal{E}_{agg}$ on $\Delta \log$ active share 0.385 (SE 0.022); 0.336 (SE 0.055) with Russell 1000/2000 switches	Table 4, col. 2, 5	consistent with one-third
Prices	per SD of elasticity, instrumented: volatility -0.383 (SE 0.123), illiquidity -0.396 (SE 0.131)	Table 5, col. 2, 8	inelastic: more volatile
Evolution	mean $\mathcal{E}_{agg}$ 0.41 to 0.27; with $\chi = 0$ the fall about twice as large	Figures 6, 8	response damped the fall

Appendix: Case Study, Limits and an Open Question

- ▣ Limits stated by the authors
 - ▶ one χ for all investors; no strategic entry, exit or fund flows (Section I.B)
 - ▶ one 13F form per institution: its active and passive funds cannot be separated (footnote 51)
 - ▶ the elasticity is for single stocks, not for the market as a whole (footnote 37)
 - ▶ those who turn passive are assumed representative of active investors (Section V.A)
- ▣ Course data: the 13F holdings are not included and public aggregates do not separate index from active funds, so the paper's passive share and elasticity path cannot be reproduced; what can be checked is on the [▶ next slide](#)
- ▣ Extension: does the pass-through change if the most elastic investors turn passive first? (Seminar 0, C3)
- ▣ Open question: does passive investing change price formation?
 - ▶ [Sammon \(2025\)](#): higher passive ownership makes prices anticipate earnings announcements less
 - ▶ passive ownership is endogenous: identification relies on index reconstitutions and flow shocks

Appendix: Case Study, What the Course Data Can Reproduce

- The 13F holdings are not in the course data, so χ cannot be re-estimated; the published arithmetic of Section V.A can be checked
 - ▶ eq. (5) with $\alpha = 70\%$ active, $\chi = 3$: $0.70 + 0.30 \times 2.1/3.1 = 0.90$; eq. (28): $1/(1 + 3 \times 0.68) = 0.33$
 - ▶ the paper's illustration: median active share 81.4% (2001) to 59.5% (2020), $\ln(81/59) = 32\%$, times 0.33: elasticity about 11% lower
- Data limitation: the passive-ownership measure of the paper cannot be rebuilt from public aggregates
 - ▶ the Financial Accounts of the United States do not separate index from active mutual funds, and their ETF series mixes passive and active funds
 - ▶ so the paper's passive-share path, and the elasticity path implied by it, are not reproduced here
- The authors also point to bank regulation, fire sales and large sovereign investors (Section VI)

Appendix: Mini-Case, Estimation and Power

- ▣ May 2003 – September 2026 (5 884 days): mean spread +0.29% a year, $t = 0.27$
- ▣ Pre-specified halves: 2003–2014 −2.28% a year ($t = -1.75$); 2015–2026 +2.85% ($t = 1.63$)
- ▣ **Minimum detectable effect (MDE)**: the smallest true effect detected with 80% power by a two-sided test at the 5% level (Normal approximation)

$$\text{MDE} = (z_{0.975} + z_{0.80}) \times \text{SE} \approx 2.8 \times \text{SE}$$

- ▶ z_p : the p -quantile of the standard Normal distribution; SE: the HAC standard error of the annualised mean spread $252 \bar{d}$, in pp a year
- ▶ full sample: 3.1% a year; 2015–2026: 4.9% a year, larger than the estimate itself
- ▣ “Not significant” is not “no effect”: under these assumptions the design does not reach 80% power for premiums below about 3.1% a year
- ▣ Causality: SPY – RSP is an outcome, not a measure of passive ownership; a causal claim needs an exogenous shifter

Appendix: A Minimal Python Workflow

```
import numpy as np, pandas as pd

URL = ('https://raw.githubusercontent.com/'
      'danpele/MFM/main/data/market/')
def load(sym, col='close'):
    d = pd.read_csv(URL + sym + '.csv',
                    index_col='date', parse_dates=True)
    return d[col].rename(sym)

spx = load('GSPC.INDX')           # index: close
spy = load('SPY.US', 'adjusted_close') # ETF
btc = load('BTC-USD.CC')          # 24/7
r_spx = np.log(spx).diff()        # one series
# joint analysis: join PRICES on common
# days first, then take returns
p = pd.concat([spx, btc], axis=1).dropna()
r = np.log(p).diff().dropna()
dd = spx / spx.cummax() - 1       # drawdown
```

- One series: returns on its own calendar, `np.log(s).diff()`
- Joint analysis: join the *prices* on common days, then take returns, so both cover the same interval (Friday → Monday)
- Returns taken before the join would pair a Bitcoin Sunday → Monday move with an S&P 500 Friday → Monday move
- Crypto trades every day: annualise with 365, not 252

Appendix: Data Checklist

- ▣ Plot the raw series and list the largest returns
- ▣ Compare two sources when possible
- ▣ Check adjustments: dividends, splits, redenominations
- ▣ Remove spurious non-trading-day records; investigate isolated outliers, and correct or remove only confirmed data errors; document every change
- ▣ For joint analyses, join the prices on common dates before computing returns; for a single asset, use its own calendar
- ▣ Watch survivorship: delisted firms vanish from today's index lists

Appendix: The Discovery Loop, Reusable Prompts (1/2)

□ Literature

- ▶ List peer-reviewed papers since 2018 on passive ownership and price informativeness. For each: DOI, identification strategy, main estimate. Flag any paper you are not sure exists.
- ▶ Resolve every DOI on Crossref; title and authors must match; read the abstract yourself

□ Hypothesis

- ▶ Give three competing explanations for SPY beating RSP after 2015 (passive flows, sector composition, earnings growth of the largest firms). For each, one testable implication the others do not share.

□ Data and code

- ▶ Write Python that reads the daily adjusted (total-return) prices of SPY and RSP from the course data, joins prices on common days, and tests H_0 : mean daily log-return spread = 0 with a Newey-West t-statistic.

Appendix: The Discovery Loop, Reusable Prompts (2/2)

▣ Robustness

- ▶ Write a pre-registered robustness plan: subsamples, HAC bandwidths, simple vs log returns, a sector-neutral version. State in advance which outcome would falsify the hypothesis.

▣ Critique

- ▶ Act as a hostile referee. List the three strongest threats to identification in this design and the test that addresses each.

▣ Report: estimate, standard error, sample, every specification tried, also the failed ones

▣ What the researcher must check: invented references (resolve each DOI); unadjusted instead of adjusted close (reproduce the total-return multiples first); returns before the join (count the common days); window shopping (fix the windows in advance); low power (report the MDE)

References (1/8)

- ▣ Acerbi, C., Tasche, D. (2002). [On the coherence of expected shortfall](#). *Journal of Banking & Finance*, 26(7), 1487–1503.
- ▣ Andersen, T.G., Bollerslev, T., Diebold, F.X., Labys, P. (2003). [Modeling and forecasting realized volatility](#). *Econometrica*, 71(2), 579–625.
- ▣ Ansari, A.F. et al. (2024). [Chronos: learning the language of time series](#). *arXiv:2403.07815*.
- ▣ Artzner, P., Delbaen, F., Eber, J.-M., Heath, D. (1999). [Coherent measures of risk](#). *Mathematical Finance*, 9(3), 203–228.
- ▣ Bachelier, L. (1900). [Théorie de la spéculation](#). *Annales scientifiques de l'École Normale Supérieure*, 17, 21–86.
- ▣ Bailey, D.H., López de Prado, M. (2014). [The deflated Sharpe ratio: correcting for selection bias, backtest overfitting, and non-normality](#). *Journal of Portfolio Management*, 40(5), 94–107.
- ▣ BCBS (1996). [Amendment to the capital accord to incorporate market risks](#). Basel Committee on Banking Supervision.
- ▣ BCBS (2019). [Minimum capital requirements for market risk](#). Basel Committee on Banking Supervision.
- ▣ Ben-David, I., Franzoni, F., Moussawi, R. (2018). [Do ETFs increase volatility?](#) *Journal of Finance*, 73(6), 2471–2535.
- ▣ BIS (2023). [Blueprint for the future monetary system: improving the old, enabling the new](#). Annual Economic Report, Chapter III.
- ▣ BIS (2025). [The next-generation monetary and financial system](#). Annual Economic Report, Chapter III.
- ▣ Black, F., Scholes, M. (1973). [The pricing of options and corporate liabilities](#). *Journal of Political Economy*, 81(3), 637–654.

References (2/8)

- ▣ Bollerslev, T. (1986). [Generalized autoregressive conditional heteroskedasticity](#). *Journal of Econometrics*, 31(3), 307–327.
- ▣ Bollerslev, T., Tauchen, G., Zhou, H. (2009). [Expected stock returns and variance risk premia](#). *Review of Financial Studies*, 22(11), 4463–4492.
- ▣ Brandimarte, P. (2018). [An Introduction to Financial Markets: A Quantitative Approach](#). Wiley.
- ▣ Brown, S.J., Goetzmann, W., Ibbotson, R.G., Ross, S.A. (1992). [Survivorship bias in performance studies](#). *Review of Financial Studies*, 5(4), 553–580.
- ▣ Budish, E., Cramton, P., Shim, J. (2015). [The high-frequency trading arms race: frequent batch auctions as a market design response](#). *Quarterly Journal of Economics*, 130(4), 1547–1621.
- ▣ Buehler, H., Gonon, L., Teichmann, J., Wood, B. (2019). [Deep hedging](#). *Quantitative Finance*, 19(8), 1271–1291.
- ▣ Burns, P., Engle, R., Mezrich, J. (1998). [Correlations and volatilities of asynchronous data](#). *Journal of Derivatives*, 5(4), 7–18.
- ▣ BVB (2017). [22 de ani de la prima tranzacție la Bursa de Valori București](#). Press release, 20 November 2017.
- ▣ BVB. [BVB indices: BET, launched 19 September 1997](#). Bucharest Stock Exchange.
- ▣ BVB. [Istoric \(History\)](#). Bursa de Valori București (the Bucharest Stock Exchange).
- ▣ Campbell, J.Y., Lo, A.W., MacKinlay, A.C. (1997). [The Econometrics of Financial Markets](#). Princeton University Press.
- ▣ Carr, P., Wu, L. (2009). [Variance risk premiums](#). *Review of Financial Studies*, 22(3), 1311–1341.
- ▣ Chakrabarti, S., Krishnakumar, R., Mazzola, G., Stamatopoulos, N., Woerner, S., Zeng, W.J. (2021). [A threshold for quantum advantage in derivative pricing](#). *Quantum*, 5, 463.

References (3/8)

- ▣ Cox, J.C., Ingersoll, J.E., Ross, S.A. (1985). A theory of the term structure of interest rates. *Econometrica*, 53(2), 385–407.
- ▣ Donnelly, C., Embrechts, P. (2010). The devil is in the tails: actuarial mathematics and the subprime mortgage crisis. *ASTIN Bulletin*, 40(1), 1–33.
- ▣ Embrechts, P., Klüppelberg, C., Mikosch, T. (1997). *Modelling Extremal Events for Insurance and Finance*. Springer.
- ▣ Engle, R.F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- ▣ Estrella, A., Mishkin, F.S. (1998). Predicting U.S. recessions: financial variables as leading indicators. *Review of Economics and Statistics*, 80(1), 45–61.
- ▣ ESMA. *Markets in Crypto-Assets Regulation (MiCA), Regulation (EU) 2023/1114*.
- ▣ Fama, E.F. (1965). The behavior of stock-market prices. *Journal of Business*, 38(1), 34–105.
- ▣ Fama, E.F. (1970). Efficient capital markets: a review of theory and empirical work. *Journal of Finance*, 25(2), 383–417.
- ▣ Fama, E.F., French, K.R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56.
- ▣ Forbes, K.J., Rigobon, R. (2002). No contagion, only interdependence: measuring stock market comovements. *Journal of Finance*, 57(5), 2223–2261.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction* (5th ed.). Springer.
- ▣ Frehen, R.G.P., Goetzmann, W.N., Rouwenhorst, K.G. (2013). New evidence on the first financial bubble. *Journal of Financial Economics*, 108(3), 585–607.
- ▣ FTSE Russell (2026). *FTSE Equity Country Classification, Interim Announcement*, April 2026.
- ▣ Garber, P.M. (1989). Tulipmania. *Journal of Political Economy*, 97(3), 535–560.

References (4/8)

- ▣ Gatheral, J., Jaisson, T., Rosenbaum, M. (2018). *Volatility is rough*. *Quantitative Finance*, 18(6), 933–949.
- ▣ Gelderblom, O. (2013). *Cities of Commerce: The Institutional Foundations of International Trade in the Low Countries, 1250–1650*. Princeton University Press.
- ▣ Gelderblom, O., Jonker, J. (2004). *Completing a financial revolution: the finance of the Dutch East India trade and the rise of the Amsterdam capital market, 1595–1612*. *Journal of Economic History*, 64(3), 641–672.
- ▣ Goldgar, A. (2007). *Tulipmania: Money, Honor, and Knowledge in the Dutch Golden Age*. University of Chicago Press.
- ▣ Gu, S., Kelly, B., Xiu, D. (2020). *Empirical asset pricing via machine learning*. *Review of Financial Studies*, 33(5), 2223–2273.
- ▣ Haddad, V., Huebner, P., Loualiche, E. (2025). *How competitive is the stock market? Theory, evidence from portfolios, and implications for the rise of passive investing*. *American Economic Review*, 115(3), 975–1018.
- ▣ Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (Eds.) (2017). *Applied Quantitative Finance* (3rd ed.). Springer.
- ▣ Härdle, W.K., Harvey, C.R., Reule, R.C.G. (2020). *Understanding cryptocurrencies*. *Journal of Financial Econometrics*, 18(2), 181–208.
- ▣ Härdle, W.K., Simar, L. (2019). *Applied Multivariate Statistical Analysis*. Springer.
- ▣ Heath, D., Jarrow, R., Morton, A. (1992). *Bond pricing and the term structure of interest rates: a new methodology for contingent claims valuation*. *Econometrica*, 60(1), 77–105.
- ▣ Herman, D., Googin, C., Liu, X., Sun, Y., Galda, A., Safro, I., Pistoia, M., Alexeev, Y. (2023). *Quantum computing for finance*. *Nature Reviews Physics*, 5(8), 450–465.
- ▣ Heston, S.L. (1993). *A closed-form solution for options with stochastic volatility with applications to bond and currency options*. *Review of Financial Studies*, 6(2), 327–343.

References (5/8)

- ▣ Hutchinson, J.M., Lo, A.W., Poggio, T. (1994). A nonparametric approach to pricing and hedging derivative securities via learning networks. *Journal of Finance*, 49(3), 851–889.
- ▣ IBM Research (2024). The 2023 IBM Research annual letter. IBM Research Blog, 2 February 2024.
- ▣ J.P. Morgan/Reuters (1996). *RiskMetrics: Technical Document* (4th ed.). New York.
- ▣ Koijen, R.S.J., Yogo, M. (2019). A demand system approach to asset pricing. *Journal of Political Economy*, 127(4), 1475–1515.
- ▣ Ledoit, O., Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5), 850–859.
- ▣ Lege asupra burselor (1929). *Lege asupra burselor* (Law on exchanges). Bucharest; digitised by Biblioteca Centrală Universitară Cluj.
- ▣ Li, D.X. (2000). On default correlation: a copula function approach. *Journal of Fixed Income*, 9(4), 43–54.
- ▣ Lintner, J. (1965). The valuation of risk assets and the selection of risky investments in stock portfolios and capital budgets. *Review of Economics and Statistics*, 47(1), 13–37.
- ▣ Lo, A.W. (2002). The statistics of Sharpe ratios. *Financial Analysts Journal*, 58(4), 36–52.
- ▣ Lo, A.W., MacKinlay, A.C. (1990). An econometric analysis of nonsynchronous trading. *Journal of Econometrics*, 45(1–2), 181–211.
- ▣ Lopez-Lira, A., Tang, Y. (2026). Can ChatGPT forecast stock price movements? Return predictability and large language models. *Journal of Financial Economics*, 184, 104335.
- ▣ Magdon-Ismail, M., Atiya, A.F., Pratap, A., Abu-Mostafa, Y.S. (2004). On the maximum drawdown of a Brownian motion. *Journal of Applied Probability*, 41(1), 147–161.
- ▣ Makarov, I., Schoar, A. (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics*, 135(2), 293–319.
- ▣ Mandelbrot, B. (1963). The variation of certain speculative prices. *Journal of Business*, 36(4), 394–419.

References (6/8)

- ▣ Markowitz, H. (1952). [Portfolio selection](#). *Journal of Finance*, 7(1), 77–91.
- ▣ McNeil, A.J., Frey, R. (2000). [Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach](#). *Journal of Empirical Finance*, 7(3–4), 271–300.
- ▣ Merton, R.C. (1973). [Theory of rational option pricing](#). *Bell Journal of Economics and Management Science*, 4(1), 141–183.
- ▣ Merton, R.C. (1980). [On estimating the expected return on the market](#). *Journal of Financial Economics*, 8(4), 323–361.
- ▣ Montanaro, A. (2015). [Quantum speedup of Monte Carlo methods](#). *Proceedings of the Royal Society A*, 471(2181), 20150301.
- ▣ Mossin, J. (1966). [Equilibrium in a capital asset market](#). *Econometrica*, 34(4), 768–783.
- ▣ Nakamoto, S. (2008). [Bitcoin: A peer-to-peer electronic cash system](#). White paper.
- ▣ NBER. [US business cycle expansions and contractions](#). National Bureau of Economic Research.
- ▣ Neal, L. (1990). *The Rise of Financial Capitalism: International Capital Markets in the Age of Reason*. Cambridge University Press.
- ▣ Opdyke, J.D. (2007). [Comparing Sharpe ratios: so where are the p-values?](#) *Journal of Asset Management*, 8(5), 308–336.
- ▣ Orús, R., Mugel, S., Lizaso, E. (2019). [Quantum computing for finance: overview and prospects](#). *Reviews in Physics*, 4, 100028.
- ▣ Palatul Bursei. [Palatul Bursei \(The Stock Exchange Palace, Bucharest\)](#). Wikipedia (Romanian), accessed 2026.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). [Are cryptos becoming alternative assets?](#). *The European Journal of Finance*, 29(10), 1064–1105.

References (7/8)

- ▣ Petajisto, A. (2017). Inefficiencies in the pricing of exchange-traded funds. *Financial Analysts Journal*, 73(1), 24–54.
- ▣ Petram, L.O. (2014). *The World's First Stock Exchange*. Columbia University Press.
- ▣ Preskill, J. (2018). Quantum computing in the NISQ era and beyond. *Quantum*, 2, 79.
- ▣ Rebentrost, P., Gupt, B., Bromley, T.R. (2018). Quantum computational finance: Monte Carlo pricing of financial derivatives. *Physical Review A*, 98(2), 022321.
- ▣ Roll, R. (1984). A simple implicit measure of the effective bid-ask spread in an efficient market. *Journal of Finance*, 39(4), 1127–1139.
- ▣ Roll, R. (1988). The international crash of October 1987. *Financial Analysts Journal*, 44(5), 19–35.
- ▣ Ross, S.A. (1976). The arbitrage theory of capital asset pricing. *Journal of Economic Theory*, 13(3), 341–360.
- ▣ Sammon, M. (2025). Passive ownership and price informativeness. *Management Science*, 71(6), 4582–4598.
- ▣ Samuelson, P.A. (1965). Rational theory of warrant pricing. *Industrial Management Review*, 6(2), 13–39. Reprinted in *Henry P. McKean Jr. Selecta* (2015), Springer, 195–232.
- ▣ Scholes, M., Williams, J. (1977). Estimating betas from nonsynchronous data. *Journal of Financial Economics*, 5(3), 309–327.
- ▣ SEC (2024). *Statement on the Approval of Spot Bitcoin Exchange-Traded Products*, 10 January 2024.
- ▣ Sharpe, W.F. (1964). Capital asset prices: a theory of market equilibrium under conditions of risk. *Journal of Finance*, 19(3), 425–442.

References (8/8)

- ▣ Stamatopoulos, N., Egger, D.J., Sun, Y., Zoufal, C., Iten, R., Shen, N., Woerner, S. (2020). Option pricing using quantum computers. *Quantum*, 4, 291.
- ▣ Sylla, R., Wright, R.E., Cowen, D.J. (2009). Alexander Hamilton, central banker: crisis management during the U.S. financial panic of 1792. *Business History Review*, 83(1), 61–86.
- ▣ Tobin, J. (1958). Liquidity preference as behavior towards risk. *Review of Economic Studies*, 25(2), 65–86.
- ▣ U.S. Congress. Guiding and Establishing National Innovation for U.S. Stablecoins Act (GENIUS Act), Public Law 119-27.
- ▣ Vasicek, O. (1977). An equilibrium characterization of the term structure. *Journal of Financial Economics*, 5(2), 177–188.
- ▣ Vaswani, A. et al. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30.
- ▣ Woerner, S., Egger, D.J. (2019). Quantum risk analysis. *npj Quantum Information*, 5, 15.
- ▣ Wu, S. et al. (2023). BloombergGPT: a large language model for finance. *arXiv:2303.17564*.
- ▣ Code and data: github.com/danpele/MFM/tree/main/Quantlets/Ch_00